

Analysis of EMI at the individual level

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```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'  
  
## The following objects are masked from 'package:stats':  
##  
##   filter, lag  
  
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
library(zoo)
```

```
##  
## Attaching package: 'zoo'  
  
## The following objects are masked from 'package:base':  
##  
##   as.Date, as.Date.numeric
```

```
library(Hmisc)
```

```
##  
## Attaching package: 'Hmisc'  
  
## The following objects are masked from 'package:dplyr':  
##  
##   src, summarize  
  
## The following objects are masked from 'package:base':  
##  
##   format.pval, units
```

```
library(lmtest)  
library(tseries)
```

```
## Registered S3 method overwritten by 'quantmod':  
##   method      from  
##   as.zoo.data.frame zoo
```

```
library(ggplot2)
library(ggrepel)
library(stargazer)
```

```
##
```

```
## Please cite as:
```

```
## Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.
```

```
## R package version 5.2.3. https://CRAN.R-project.org/package=stargazer
```

```
library(car)
```

```
## Loading required package: carData
```

```
##
```

```
## Attaching package: 'car'
```

```
## The following object is masked from 'package:dplyr':
```

```
##
```

```
##      recode
```

```
library(sandwich)
library(clubSandwich)
```

```
## Registered S3 method overwritten by 'clubSandwich':
```

```
##   method      from
```

```
##   bread.mlm sandwich
```

```
library(lme4)
```

```
## Loading required package: Matrix
```

```
library(lmerTest)
```

```
##
```

```
## Attaching package: 'lmerTest'
```

```
## The following object is masked from 'package:lme4':
```

```
##
```

```
##      lmer
```

```
## The following object is masked from 'package:stats':
```

```
##
```

```
##      step
```

```
library(ggeffects)
library(margins)
library(marginaleffects)
```

```
##
## Attaching package: 'marginaleffects'

## The following object is masked from 'package:lme4':
##
##      refit
```

```
library(boot)
```

```
##
## Attaching package: 'boot'

## The following object is masked from 'package:car':
##
##      logit
```

```
library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'

## The following object is masked from 'package:dplyr':
##
##      group_rows
```

```
library(modelsummary)
```

```
##
## Attaching package: 'modelsummary'

## The following object is masked from 'package:Hmisc':
##
##      Mean
```

```
library(visreg)
library(tinytable)
```

```
set.seed(1234)
setwd(".")
```

```
output_path <- "../output"
f <- function(x) format(x, digits = 3, nsmall = 2)
f_est <- function(x) formatC(x, format="f", digits=3)
f_p <- function(x) {
```

```

    ifelse(
      x < 0.001,
      formatC(x, format = "e", digits = 3),
      formatC(x, format = "f", digits = 3)
    )
  }

df <- read.csv("../data/memberlevel_sbertllms.csv")
df$emi <- df$rating_emb_pool_z
df$ideology <- abs(df$nominate_dim1)

df$bioguide_id <- factor(df$bioguide_id)
df$congress <- factor(df$congress)
df$republican <- ifelse(
  df$party_name == "Republican Party",
  1,
  0
)

df$female <- ifelse(
  df$gender == "F",
  1,
  0
)

df$senate <- ifelse(
  df$chamber == "Senate",
  1,
  0
)

cor.test(df$emi, df$nominate_dim1_absfloordistance)

##
## Pearson's product-moment correlation
##
## data: df$emi and df$nominate_dim1_absfloordistance
## t = -23.76, df = 37684, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.1314231 -0.1115286
## sample estimates:
## cor
## -0.1214881

cor.test(df$emi, df$ideology)

##
## Pearson's product-moment correlation
##
## data: df$emi and df$ideology

```

```
## t = -15.556, df = 37684, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.08990029 -0.06983659
## sample estimates:
##      cor
## -0.07987653
```

```
vars <- c(
  "emi", "nominate_dim1_absfloordistance", "congress",
  "gender", "chamber_opposition", "bioguide_id",
  "republican", "senate", "ideology",
  grep("^prop_topic_", names(df), value = TRUE)
)

df <- df[complete.cases(df[, vars]), ]

vars_to_scale <- c(
  "emi",
  "nominate_dim1_absfloordistance",
  "ideology",
  grep("^prop_topic_", names(df), value = TRUE)
)

df <- df |>
  mutate(
    across(
      all_of(vars_to_scale),
      ~ as.numeric(scale(.x))
    )
  )

emi_model_base <- lmer(emi ~ 1 + ideology + congress
  + republican
  + senate
  + (1|bioguide_id),
  data = df )
summary(emi_model_base)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## emi ~ 1 + ideology + congress + republican + senate + (1 | bioguide_id)
## Data: df
##
## REML criterion at convergence: 92362.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -11.5509  -0.4521   0.0017   0.4652   9.7865
##
## Random effects:
## Groups      Name             Variance Std.Dev.
## bioguide_id (Intercept) 0.3972    0.6302
```

```

## Residual              0.5083    0.7130
## Number of obs: 37686, groups:  bioguide_id, 7876
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept) -1.045e+00  4.951e-02  2.893e+04 -21.112 < 2e-16 ***
## ideology    -1.158e-01  8.850e-03  6.708e+03 -13.087 < 2e-16 ***
## congress44   2.703e-01  6.151e-02  3.747e+04  4.394 1.12e-05 ***
## congress45   3.310e-01  6.210e-02  3.757e+04  5.331 9.84e-08 ***
## congress46   6.462e-01  6.333e-02  3.734e+04 10.203 < 2e-16 ***
## congress47   6.156e-01  6.377e-02  3.732e+04  9.653 < 2e-16 ***
## congress48   9.934e-01  6.348e-02  3.675e+04 15.650 < 2e-16 ***
## congress49   9.307e-01  6.294e-02  3.665e+04 14.786 < 2e-16 ***
## congress50   7.138e-01  6.358e-02  3.669e+04 11.227 < 2e-16 ***
## congress51   5.432e-01  6.268e-02  3.618e+04  8.665 < 2e-16 ***
## congress52   7.125e-01  6.351e-02  3.587e+04 11.217 < 2e-16 ***
## congress53   4.761e-01  6.268e-02  3.569e+04  7.596 3.14e-14 ***
## congress54   7.779e-01  6.330e-02  3.546e+04 12.289 < 2e-16 ***
## congress55   6.726e-01  6.324e-02  3.590e+04 10.635 < 2e-16 ***
## congress56   7.049e-01  6.315e-02  3.582e+04 11.164 < 2e-16 ***
## congress57   6.216e-01  6.283e-02  3.585e+04  9.893 < 2e-16 ***
## congress58   8.673e-01  6.188e-02  3.556e+04 14.014 < 2e-16 ***
## congress59   7.998e-01  6.191e-02  3.554e+04 12.920 < 2e-16 ***
## congress60   9.712e-01  6.147e-02  3.535e+04 15.799 < 2e-16 ***
## congress61   7.571e-01  6.128e-02  3.513e+04 12.354 < 2e-16 ***
## congress62   6.389e-01  6.095e-02  3.466e+04 10.483 < 2e-16 ***
## congress63   5.719e-01  6.049e-02  3.428e+04  9.455 < 2e-16 ***
## congress64   5.486e-01  6.053e-02  3.477e+04  9.064 < 2e-16 ***
## congress65   3.464e-01  6.011e-02  3.451e+04  5.763 8.35e-09 ***
## congress66   4.569e-01  6.009e-02  3.440e+04  7.604 2.95e-14 ***
## congress67   3.740e-01  6.017e-02  3.424e+04  6.217 5.14e-10 ***
## congress68   4.553e-01  6.042e-02  3.450e+04  7.535 5.02e-14 ***
## congress69   6.207e-01  6.045e-02  3.454e+04 10.268 < 2e-16 ***
## congress70   7.284e-01  6.049e-02  3.457e+04 12.042 < 2e-16 ***
## congress71   6.588e-01  6.006e-02  3.415e+04 10.969 < 2e-16 ***
## congress72   5.749e-01  6.037e-02  3.440e+04  9.524 < 2e-16 ***
## congress73   5.791e-01  6.039e-02  3.427e+04  9.590 < 2e-16 ***
## congress74   6.867e-01  6.051e-02  3.439e+04 11.348 < 2e-16 ***
## congress75   6.953e-01  6.101e-02  3.462e+04 11.397 < 2e-16 ***
## congress76   6.777e-01  6.026e-02  3.437e+04 11.247 < 2e-16 ***
## congress77   7.759e-01  6.031e-02  3.447e+04 12.864 < 2e-16 ***
## congress78   7.202e-01  6.010e-02  3.433e+04 11.985 < 2e-16 ***
## congress79   8.026e-01  6.020e-02  3.433e+04 13.331 < 2e-16 ***
## congress80   7.246e-01  6.012e-02  3.422e+04 12.052 < 2e-16 ***
## congress81   7.063e-01  6.002e-02  3.420e+04 11.768 < 2e-16 ***
## congress82   7.222e-01  5.997e-02  3.427e+04 12.043 < 2e-16 ***
## congress83   8.451e-01  5.996e-02  3.426e+04 14.094 < 2e-16 ***
## congress84   9.180e-01  6.000e-02  3.435e+04 15.300 < 2e-16 ***
## congress85   9.030e-01  5.980e-02  3.426e+04 15.099 < 2e-16 ***
## congress86   9.342e-01  5.978e-02  3.418e+04 15.627 < 2e-16 ***
## congress87   9.907e-01  5.962e-02  3.418e+04 16.619 < 2e-16 ***
## congress88   9.060e-01  5.974e-02  3.428e+04 15.166 < 2e-16 ***
## congress89   1.226e+00  5.986e-02  3.419e+04 20.480 < 2e-16 ***
## congress90   1.128e+00  5.981e-02  3.435e+04 18.867 < 2e-16 ***

```

```
## congress91 1.154e+00 5.964e-02 3.422e+04 19.343 < 2e-16 ***
## congress92 1.288e+00 5.979e-02 3.424e+04 21.540 < 2e-16 ***
## congress93 1.398e+00 5.985e-02 3.427e+04 23.364 < 2e-16 ***
## congress94 1.483e+00 5.983e-02 3.439e+04 24.789 < 2e-16 ***
## congress95 1.502e+00 5.979e-02 3.423e+04 25.115 < 2e-16 ***
## congress96 1.339e+00 5.982e-02 3.424e+04 22.383 < 2e-16 ***
## congress97 1.177e+00 5.987e-02 3.412e+04 19.659 < 2e-16 ***
## congress98 1.330e+00 5.980e-02 3.420e+04 22.245 < 2e-16 ***
## congress99 1.250e+00 5.981e-02 3.415e+04 20.908 < 2e-16 ***
## congress100 1.352e+00 5.974e-02 3.414e+04 22.628 < 2e-16 ***
## congress101 1.341e+00 5.973e-02 3.409e+04 22.452 < 2e-16 ***
## congress102 1.377e+00 5.969e-02 3.406e+04 23.069 < 2e-16 ***
## congress103 1.142e+00 5.973e-02 3.394e+04 19.114 < 2e-16 ***
## congress104 1.022e+00 5.977e-02 3.386e+04 17.091 < 2e-16 ***
## congress105 1.280e+00 5.978e-02 3.378e+04 21.417 < 2e-16 ***
## congress106 1.520e+00 5.992e-02 3.384e+04 25.372 < 2e-16 ***
## congress107 1.422e+00 5.994e-02 3.377e+04 23.729 < 2e-16 ***
## congress108 1.473e+00 6.000e-02 3.372e+04 24.552 < 2e-16 ***
## congress109 1.575e+00 5.998e-02 3.367e+04 26.258 < 2e-16 ***
## congress110 1.731e+00 5.983e-02 3.348e+04 28.930 < 2e-16 ***
## congress111 1.441e+00 5.987e-02 3.331e+04 24.069 < 2e-16 ***
## congress112 1.277e+00 6.012e-02 3.335e+04 21.232 < 2e-16 ***
## congress113 1.468e+00 6.015e-02 3.312e+04 24.405 < 2e-16 ***
## congress114 1.621e+00 6.038e-02 3.299e+04 26.854 < 2e-16 ***
## congress115 1.856e+00 6.034e-02 3.273e+04 30.758 < 2e-16 ***
## congress116 1.763e+00 6.058e-02 3.242e+04 29.096 < 2e-16 ***
## congress117 1.482e+00 6.068e-02 3.211e+04 24.424 < 2e-16 ***
## congress118 1.449e+00 6.100e-02 3.161e+04 23.744 < 2e-16 ***
## republican 6.273e-02 1.732e-02 6.411e+03 3.622 0.000295 ***
## senate -6.496e-02 1.780e-02 1.843e+04 -3.649 0.000264 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation matrix not shown by default, as p = 79 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)          if you need it
```

```
vif(emi_model_base)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## ideology    1.109646 1          1.053397
## congress    1.101391 75          1.000644
## republican  1.055078 1          1.027170
## senate      1.007450 1          1.003718
```

```
emi_model <- lmer(emi ~ 1 + ideology + congress
+ chamber_opposition + president_opposition + republican
+ female + senate
+ prop_topic_Agriculture + prop_topic_Civil.Rights
+ prop_topic_Defense + prop_topic_Domestic.Commerce
+ prop_topic_Education + prop_topic_Energy
+ prop_topic_Environment + prop_topic_Foreign.Trade
```

```

+ prop_topic_Government.Operations + prop_topic_Health
+ prop_topic_Housing + prop_topic_Immigration
+ prop_topic_International.Affairs + prop_topic_Labor
+ prop_topic_Law.and.Crime + prop_topic_Macroeconomics
+ prop_topic_Public.Lands + prop_topic_Social.Welfare
+ prop_topic_Technology + prop_topic_Transportation
+ (1|bioguide_id),
data = df )
summary(emi_model)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## emi ~ 1 + ideology + congress + chamber_opposition + president_opposition +
## republican + female + senate + prop_topic_Agriculture + prop_topic_Civil.Rights +
## prop_topic_Defense + prop_topic_Domestic.Commerce + prop_topic_Education +
## prop_topic_Energy + prop_topic_Environment + prop_topic_Foreign.Trade +
## prop_topic_Government.Operations + prop_topic_Health + prop_topic_Housing +
## prop_topic_Immigration + prop_topic_International.Affairs +
## prop_topic_Labor + prop_topic_Law.and.Crime + prop_topic_Macroeconomics +
## prop_topic_Public.Lands + prop_topic_Social.Welfare + prop_topic_Technology +
## prop_topic_Transportation + (1 | bioguide_id)
## Data: df
##
## REML criterion at convergence: 84146.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -10.8461  -0.4425   0.0043   0.4432  10.6106
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## bioguide_id (Intercept) 0.2536   0.5036
## Residual              0.4217   0.6494
## Number of obs: 37686, groups: bioguide_id, 7876
##
## Fixed effects:
##
##              Estimate Std. Error      df t value
## (Intercept)  -8.158e-01  4.426e-02 2.932e+04 -18.435
## ideology      -7.174e-02  7.422e-03 5.987e+03  -9.665
## congress44     3.123e-01  5.519e-02 3.745e+04   5.658
## congress45     3.685e-01  5.557e-02 3.752e+04   6.632
## congress46     6.111e-01  5.653e-02 3.722e+04  10.811
## congress47     5.635e-01  5.700e-02 3.721e+04   9.887
## congress48     8.593e-01  5.662e-02 3.659e+04  15.178
## congress49     8.234e-01  5.612e-02 3.649e+04  14.671
## congress50     6.827e-01  5.675e-02 3.657e+04  12.032
## congress51     5.541e-01  5.579e-02 3.602e+04   9.933
## congress52     7.243e-01  5.654e-02 3.574e+04  12.811
## congress53     4.872e-01  5.577e-02 3.553e+04   8.736
## congress54     7.499e-01  5.632e-02 3.537e+04  13.315
## congress55     6.860e-01  5.628e-02 3.579e+04  12.189
## congress56     6.783e-01  5.619e-02 3.573e+04  12.072

```

## congress57	5.974e-01	5.597e-02	3.579e+04	10.673
## congress58	7.661e-01	5.507e-02	3.552e+04	13.910
## congress59	6.607e-01	5.523e-02	3.554e+04	11.963
## congress60	8.250e-01	5.471e-02	3.534e+04	15.080
## congress61	6.371e-01	5.471e-02	3.520e+04	11.644
## congress62	5.842e-01	5.425e-02	3.470e+04	10.769
## congress63	4.686e-01	5.378e-02	3.430e+04	8.712
## congress64	3.929e-01	5.390e-02	3.484e+04	7.290
## congress65	2.818e-01	5.359e-02	3.460e+04	5.258
## congress66	3.117e-01	5.360e-02	3.452e+04	5.815
## congress67	2.705e-01	5.369e-02	3.438e+04	5.038
## congress68	4.619e-01	5.385e-02	3.462e+04	8.578
## congress69	5.269e-01	5.388e-02	3.468e+04	9.779
## congress70	6.119e-01	5.388e-02	3.471e+04	11.356
## congress71	5.654e-01	5.366e-02	3.433e+04	10.537
## congress72	6.140e-01	5.384e-02	3.456e+04	11.403
## congress73	5.194e-01	5.390e-02	3.439e+04	9.635
## congress74	5.963e-01	5.405e-02	3.451e+04	11.032
## congress75	6.091e-01	5.459e-02	3.478e+04	11.157
## congress76	6.522e-01	5.396e-02	3.454e+04	12.087
## congress77	7.481e-01	5.412e-02	3.466e+04	13.823
## congress78	7.131e-01	5.384e-02	3.450e+04	13.244
## congress79	7.476e-01	5.407e-02	3.456e+04	13.828
## congress80	6.797e-01	5.399e-02	3.449e+04	12.590
## congress81	6.867e-01	5.383e-02	3.441e+04	12.757
## congress82	6.739e-01	5.368e-02	3.445e+04	12.555
## congress83	7.674e-01	5.379e-02	3.453e+04	14.266
## congress84	7.827e-01	5.387e-02	3.462e+04	14.530
## congress85	8.533e-01	5.362e-02	3.451e+04	15.914
## congress86	8.833e-01	5.378e-02	3.451e+04	16.425
## congress87	9.070e-01	5.372e-02	3.451e+04	16.884
## congress88	8.993e-01	5.381e-02	3.459e+04	16.714
## congress89	1.101e+00	5.405e-02	3.458e+04	20.370
## congress90	1.043e+00	5.389e-02	3.472e+04	19.346
## congress91	1.019e+00	5.371e-02	3.459e+04	18.976
## congress92	1.086e+00	5.400e-02	3.467e+04	20.116
## congress93	1.195e+00	5.398e-02	3.470e+04	22.140
## congress94	1.281e+00	5.420e-02	3.490e+04	23.627
## congress95	1.294e+00	5.408e-02	3.470e+04	23.936
## congress96	1.162e+00	5.420e-02	3.472e+04	21.445
## congress97	1.056e+00	5.433e-02	3.463e+04	19.438
## congress98	1.156e+00	5.437e-02	3.477e+04	21.267
## congress99	1.094e+00	5.432e-02	3.470e+04	20.147
## congress100	1.103e+00	5.422e-02	3.470e+04	20.335
## congress101	1.125e+00	5.444e-02	3.474e+04	20.663
## congress102	1.135e+00	5.417e-02	3.461e+04	20.952
## congress103	9.462e-01	5.419e-02	3.446e+04	17.459
## congress104	8.809e-01	5.457e-02	3.452e+04	16.143
## congress105	1.080e+00	5.417e-02	3.432e+04	19.941
## congress106	1.223e+00	5.468e-02	3.448e+04	22.363
## congress107	1.219e+00	5.445e-02	3.434e+04	22.378
## congress108	1.194e+00	5.460e-02	3.428e+04	21.867
## congress109	1.281e+00	5.456e-02	3.422e+04	23.480
## congress110	1.359e+00	5.454e-02	3.409e+04	24.912

## congress111	1.214e+00	5.495e-02	3.401e+04	22.103
## congress112	1.076e+00	5.511e-02	3.405e+04	19.518
## congress113	1.218e+00	5.479e-02	3.371e+04	22.233
## congress114	1.241e+00	5.504e-02	3.356e+04	22.553
## congress115	1.575e+00	5.504e-02	3.323e+04	28.612
## congress116	1.587e+00	5.503e-02	3.290e+04	28.844
## congress117	1.371e+00	5.516e-02	3.262e+04	24.862
## congress118	1.274e+00	5.555e-02	3.215e+04	22.934
## chamber_opposition	-1.388e-01	8.697e-03	3.597e+04	-15.958
## president_opposition	-1.641e-02	8.222e-03	3.381e+04	-1.996
## republican	1.587e-02	1.452e-02	5.680e+03	1.093
## female	8.505e-03	3.406e-02	5.407e+03	0.250
## senate	-8.780e-02	1.561e-02	1.497e+04	-5.625
## prop_topic_Agriculture	1.866e-01	5.291e-03	3.733e+04	35.263
## prop_topic_Civil.Rights	4.745e-02	4.677e-03	3.754e+04	10.147
## prop_topic_Defense	3.490e-01	5.746e-03	3.737e+04	60.745
## prop_topic_Domestic.Commerce	1.431e-01	4.535e-03	3.756e+04	31.559
## prop_topic_Education	1.492e-01	4.400e-03	3.680e+04	33.907
## prop_topic_Energy	1.687e-01	4.602e-03	3.703e+04	36.655
## prop_topic_Environment	1.376e-01	4.582e-03	3.660e+04	30.030
## prop_topic_Foreign.Trade	9.877e-02	4.251e-03	3.682e+04	23.234
## prop_topic_Government.Operations	2.886e-01	6.912e-03	3.758e+04	41.746
## prop_topic_Health	1.771e-01	5.066e-03	3.752e+04	34.954
## prop_topic_Housing	7.722e-02	4.188e-03	3.688e+04	18.437
## prop_topic_Immigration	3.255e-02	4.192e-03	3.734e+04	7.763
## prop_topic_International.Affairs	1.343e-01	5.326e-03	3.758e+04	25.217
## prop_topic_Labor	9.302e-02	4.200e-03	3.664e+04	22.145
## prop_topic_Law.and.Crime	1.736e-01	4.659e-03	3.748e+04	37.265
## prop_topic_Macroeconomics	8.248e-02	4.976e-03	3.731e+04	16.575
## prop_topic_Public.Lands	3.656e-01	5.299e-03	3.707e+04	69.004
## prop_topic_Social.Welfare	6.252e-02	4.290e-03	3.603e+04	14.574
## prop_topic_Technology	1.542e-01	4.332e-03	3.693e+04	35.590
## prop_topic_Transportation	2.344e-01	4.573e-03	3.752e+04	51.267
##	Pr(> t)			
## (Intercept)	< 2e-16 ***			
## ideology	< 2e-16 ***			
## congress44	1.54e-08 ***			
## congress45	3.36e-11 ***			
## congress46	< 2e-16 ***			
## congress47	< 2e-16 ***			
## congress48	< 2e-16 ***			
## congress49	< 2e-16 ***			
## congress50	< 2e-16 ***			
## congress51	< 2e-16 ***			
## congress52	< 2e-16 ***			
## congress53	< 2e-16 ***			
## congress54	< 2e-16 ***			
## congress55	< 2e-16 ***			
## congress56	< 2e-16 ***			
## congress57	< 2e-16 ***			
## congress58	< 2e-16 ***			
## congress59	< 2e-16 ***			
## congress60	< 2e-16 ***			
## congress61	< 2e-16 ***			

## congress62	< 2e-16 ***
## congress63	< 2e-16 ***
## congress64	3.17e-13 ***
## congress65	1.47e-07 ***
## congress66	6.10e-09 ***
## congress67	4.73e-07 ***
## congress68	< 2e-16 ***
## congress69	< 2e-16 ***
## congress70	< 2e-16 ***
## congress71	< 2e-16 ***
## congress72	< 2e-16 ***
## congress73	< 2e-16 ***
## congress74	< 2e-16 ***
## congress75	< 2e-16 ***
## congress76	< 2e-16 ***
## congress77	< 2e-16 ***
## congress78	< 2e-16 ***
## congress79	< 2e-16 ***
## congress80	< 2e-16 ***
## congress81	< 2e-16 ***
## congress82	< 2e-16 ***
## congress83	< 2e-16 ***
## congress84	< 2e-16 ***
## congress85	< 2e-16 ***
## congress86	< 2e-16 ***
## congress87	< 2e-16 ***
## congress88	< 2e-16 ***
## congress89	< 2e-16 ***
## congress90	< 2e-16 ***
## congress91	< 2e-16 ***
## congress92	< 2e-16 ***
## congress93	< 2e-16 ***
## congress94	< 2e-16 ***
## congress95	< 2e-16 ***
## congress96	< 2e-16 ***
## congress97	< 2e-16 ***
## congress98	< 2e-16 ***
## congress99	< 2e-16 ***
## congress100	< 2e-16 ***
## congress101	< 2e-16 ***
## congress102	< 2e-16 ***
## congress103	< 2e-16 ***
## congress104	< 2e-16 ***
## congress105	< 2e-16 ***
## congress106	< 2e-16 ***
## congress107	< 2e-16 ***
## congress108	< 2e-16 ***
## congress109	< 2e-16 ***
## congress110	< 2e-16 ***
## congress111	< 2e-16 ***
## congress112	< 2e-16 ***
## congress113	< 2e-16 ***
## congress114	< 2e-16 ***
## congress115	< 2e-16 ***

```

## congress116          < 2e-16 ***
## congress117          < 2e-16 ***
## congress118          < 2e-16 ***
## chamber_opposition   < 2e-16 ***
## president_opposition 0.0459 *
## republican           0.2744
## female               0.8028
## senate               1.89e-08 ***
## prop_topic_Agriculture < 2e-16 ***
## prop_topic_Civil.Rights < 2e-16 ***
## prop_topic_Defense    < 2e-16 ***
## prop_topic_Domestic.Commerce < 2e-16 ***
## prop_topic_Education  < 2e-16 ***
## prop_topic_Energy     < 2e-16 ***
## prop_topic_Environment < 2e-16 ***
## prop_topic_Foreign.Trade < 2e-16 ***
## prop_topic_Government.Operations < 2e-16 ***
## prop_topic_Health     < 2e-16 ***
## prop_topic_Housing    < 2e-16 ***
## prop_topic_Immigration 8.50e-15 ***
## prop_topic_International.Affairs < 2e-16 ***
## prop_topic_Labor      < 2e-16 ***
## prop_topic_Law.and.Crime < 2e-16 ***
## prop_topic_Macroeconomics < 2e-16 ***
## prop_topic_Public.Lands < 2e-16 ***
## prop_topic_Social.Welfare < 2e-16 ***
## prop_topic_Technology  < 2e-16 ***
## prop_topic_Transportation < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

##
## Correlation matrix not shown by default, as p = 102 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)      if you need it

```

```
vif(emi_model)
```

```

##              GVIF Df GVIF^(1/(2*Df))
## ideology      1.127080 1      1.061640
## congress      3.777866 75      1.008900
## chamber_opposition 1.187433 1      1.089694
## president_opposition 1.207182 1      1.098718
## republican    1.070722 1      1.034757
## female        1.140852 1      1.068107
## senate        1.037682 1      1.018667
## prop_topic_Agriculture 1.427481 1      1.194772
## prop_topic_Civil.Rights 1.232029 1      1.109968
## prop_topic_Defense    1.755856 1      1.325087
## prop_topic_Domestic.Commerce 1.246271 1      1.116365
## prop_topic_Education  1.149099 1      1.071960
## prop_topic_Energy     1.289667 1      1.135635
## prop_topic_Environment 1.169433 1      1.081403

```

```
## prop_topic_Foreign.Trade      1.194033  1      1.092718
## prop_topic_Government.Operations 2.569162  1      1.602861
## prop_topic_Health            1.321938  1      1.149755
## prop_topic_Housing           1.084462  1      1.041375
## prop_topic_Immigration        1.070810  1      1.034799
## prop_topic_International.Affairs 1.416649  1      1.190231
## prop_topic_Labor              1.128277  1      1.062204
## prop_topic_Law.and.Crime      1.327129  1      1.152011
## prop_topic_Macroeconomics     1.410232  1      1.187532
## prop_topic_Public.Lands       1.491573  1      1.221300
## prop_topic_Social.Welfare     1.119815  1      1.058213
## prop_topic_Technology         1.115194  1      1.056027
## prop_topic_Transportation     1.257751  1      1.121495
```

```
vars_morecontrols <- c(
  "emi", "nominate_dim1_absfloordistance", "congress", "ideology",
  "gender", "chamber_opposition", "bioguide_id", "republican", "senate",
  "seniority_terms", "first_term",
  "committee_chair", "subcommittee_chair",
  "majority_party_leadership", "minority_party_leadership",
  "vote_share_lastelection", "served_instateleg", "log_les",
  grep("^prop_topic_", names(df), value = TRUE)
)

df_morecontrols1973 <- df[complete.cases(df[, vars_morecontrols]), ]
# "log_speechcount", "log_tokencount",
vars_to_scale <- c(
  "emi",
  "nominate_dim1_absfloordistance",
  "ideology",
  "seniority_terms",
  "vote_share_lastelection",
  "log_les",
  grep("^prop_topic_", names(df), value = TRUE)
)

df_morecontrols1973 <- df_morecontrols1973 |>
  mutate(
    across(
      all_of(vars_to_scale),
      ~ as.numeric(scale(.x))
    )
  )

emi_model_morecontrols1973 <- lmer(emi ~ 1 + ideology + congress
  + chamber_opposition + president_opposition + republican
  + female + senate
  + seniority_terms + first_term
  + committee_chair + subcommittee_chair
  + majority_party_leadership + minority_party_leadership
  + vote_share_lastelection + served_instateleg
  + prop_topic_Agriculture + prop_topic_Civil.Rights
  + prop_topic_Defense + prop_topic_Domestic.Commerce
  + prop_topic_Education + prop_topic_Energy
```

```

+ prop_topic_Environment + prop_topic_Foreign.Trade
+ prop_topic_Government.Operations + prop_topic_Health
+ prop_topic_Housing + prop_topic_Immigration
+ prop_topic_International.Affairs + prop_topic_Labor
+ prop_topic_Law.and.Crime + prop_topic_Macroeconomics
+ prop_topic_Public.Lands + prop_topic_Social.Welfare
+ prop_topic_Technology + prop_topic_Transportation
+ (1|bioguide_id),
data = df_morecontrols1973)

summary(emi_model_morecontrols1973)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## emi ~ 1 + ideology + congress + chamber_opposition + president_opposition +
## republican + female + senate + seniority_terms + first_term +
## committee_chair + subcommittee_chair + majority_party_leadership +
## minority_party_leadership + vote_share_lastelection + served_instateleg +
## prop_topic_Agriculture + prop_topic_Civil.Rights + prop_topic_Defense +
## prop_topic_Domestic.Commerce + prop_topic_Education + prop_topic_Energy +
## prop_topic_Environment + prop_topic_Foreign.Trade + prop_topic_Government.Operations +
## prop_topic_Health + prop_topic_Housing + prop_topic_Immigration +
## prop_topic_International.Affairs + prop_topic_Labor + prop_topic_Law.and.Crime +
## prop_topic_Macroeconomics + prop_topic_Public.Lands + prop_topic_Social.Welfare +
## prop_topic_Technology + prop_topic_Transportation + (1 | bioguide_id)
## Data: df_morecontrols1973
##
## REML criterion at convergence: 28420.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -8.6563 -0.5335  0.0019  0.5389 10.3226
##
## Random effects:
##   Groups      Name      Variance Std.Dev.
## bioguide_id (Intercept) 0.2295   0.4791
## Residual                0.3583   0.5986
## Number of obs: 13682, groups: bioguide_id, 2454
##
## Fixed effects:
##
##              Estimate Std. Error      df t value
## (Intercept)   1.945e-02  3.811e-02  8.334e+03  0.510
## ideology      -1.477e-01  1.193e-02  2.635e+03 -12.378
## congress94     1.205e-01  3.840e-02  1.217e+04  3.139
## congress95     1.122e-01  3.957e-02  1.270e+04  2.836
## congress96    -6.284e-02  4.003e-02  1.298e+04 -1.570
## congress97    -1.791e-01  4.122e-02  1.322e+04 -4.345
## congress98    -6.339e-02  4.161e-02  1.335e+04 -1.523
## congress99    -1.165e-01  4.195e-02  1.344e+04 -2.776
## congress100   -1.137e-01  4.194e-02  1.351e+04 -2.711
## congress101   -6.871e-02  4.245e-02  1.357e+04 -1.618
## congress102   -8.922e-02  4.247e-02  1.361e+04 -2.101

```

## congress103	-3.001e-01	4.342e-02	1.362e+04	-6.912
## congress104	-3.837e-01	4.342e-02	1.359e+04	-8.838
## congress105	-1.654e-01	4.363e-02	1.351e+04	-3.791
## congress106	2.119e-02	4.416e-02	1.341e+04	0.480
## congress107	-5.319e-03	4.424e-02	1.326e+04	-0.120
## congress108	-3.045e-02	4.492e-02	1.308e+04	-0.678
## congress109	8.498e-02	4.499e-02	1.284e+04	1.889
## congress110	1.594e-01	4.528e-02	1.254e+04	3.520
## congress111	-2.848e-02	4.640e-02	1.237e+04	-0.614
## congress112	-1.743e-01	4.641e-02	1.205e+04	-3.756
## congress113	-1.573e-02	4.620e-02	1.159e+04	-0.341
## congress114	3.097e-02	4.710e-02	1.125e+04	0.658
## congress115	4.483e-01	4.752e-02	1.082e+04	9.433
## congress116	4.142e-01	4.769e-02	1.030e+04	8.685
## congress117	1.821e-01	4.833e-02	9.900e+03	3.769
## congress118	6.179e-02	4.826e-02	9.412e+03	1.280
## chamber_opposition	-1.053e-01	1.213e-02	1.225e+04	-8.684
## president_opposition	-8.024e-02	1.154e-02	1.238e+04	-6.954
## republican	-6.175e-02	2.434e-02	2.462e+03	-2.537
## female	5.807e-02	3.632e-02	2.403e+03	1.599
## senate	-6.284e-02	2.639e-02	6.710e+03	-2.381
## seniority_terms	4.888e-03	9.149e-03	9.720e+03	0.534
## first_term	2.291e-02	1.768e-02	1.253e+04	1.296
## committee_chair	3.714e-01	2.473e-02	1.256e+04	15.018
## subcommittee_chair	2.398e-01	1.471e-02	1.301e+04	16.304
## majority_party_leadership	-1.663e-02	4.452e-02	1.238e+04	-0.374
## minority_party_leadership	1.326e-01	4.474e-02	1.230e+04	2.963
## vote_share_lastelection	1.579e-02	7.414e-03	1.362e+04	2.129
## served_instateleg	-6.565e-03	2.334e-02	2.182e+03	-0.281
## prop_topic_Agriculture	1.681e-01	8.445e-03	1.327e+04	19.904
## prop_topic_Civil.Rights	7.689e-02	8.038e-03	1.362e+04	9.565
## prop_topic_Defense	3.972e-01	1.055e-02	1.358e+04	37.651
## prop_topic_Domestic.Commerce	2.107e-01	7.789e-03	1.361e+04	27.048
## prop_topic_Education	2.440e-01	7.436e-03	1.360e+04	32.814
## prop_topic_Energy	2.436e-01	8.201e-03	1.362e+04	29.696
## prop_topic_Environment	2.227e-01	7.464e-03	1.356e+04	29.832
## prop_topic_Foreign.Trade	9.491e-02	6.741e-03	1.337e+04	14.079
## prop_topic_Government.Operations	2.573e-01	1.081e-02	1.357e+04	23.808
## prop_topic_Health	2.827e-01	9.046e-03	1.362e+04	31.250
## prop_topic_Housing	8.105e-02	6.834e-03	1.361e+04	11.860
## prop_topic_Immigration	2.718e-02	6.744e-03	1.355e+04	4.030
## prop_topic_International.Affairs	1.537e-01	9.704e-03	1.347e+04	15.844
## prop_topic_Labor	1.037e-01	6.820e-03	1.353e+04	15.203
## prop_topic_Law.and.Crime	1.377e-01	7.988e-03	1.361e+04	17.235
## prop_topic_Macroeconomics	8.015e-02	8.964e-03	1.359e+04	8.941
## prop_topic_Public.Lands	3.090e-01	8.476e-03	1.339e+04	36.455
## prop_topic_Social.Welfare	1.063e-01	6.763e-03	1.334e+04	15.718
## prop_topic_Technology	2.470e-01	7.146e-03	1.358e+04	34.571
## prop_topic_Transportation	2.434e-01	7.832e-03	1.361e+04	31.084
##	Pr(> t)			
## (Intercept)	0.609789			
## ideology	< 2e-16 ***			
## congress94	0.001702 **			
## congress95	0.004572 **			

## congress96	0.116479	
## congress97	1.40e-05	***
## congress98	0.127679	
## congress99	0.005506	**
## congress100	0.006714	**
## congress101	0.105599	
## congress102	0.035677	*
## congress103	5.00e-12	***
## congress104	< 2e-16	***
## congress105	0.000151	***
## congress106	0.631366	
## congress107	0.904309	
## congress108	0.497815	
## congress109	0.058951	.
## congress110	0.000433	***
## congress111	0.539412	
## congress112	0.000173	***
## congress113	0.733476	
## congress114	0.510795	
## congress115	< 2e-16	***
## congress116	< 2e-16	***
## congress117	0.000165	***
## congress118	0.200458	
## chamber_opposition	< 2e-16	***
## president_opposition	3.72e-12	***
## republican	0.011240	*
## female	0.110023	
## senate	0.017299	*
## seniority_terms	0.593159	
## first_term	0.195119	
## committee_chair	< 2e-16	***
## subcommittee_chair	< 2e-16	***
## majority_party_leadership	0.708678	
## minority_party_leadership	0.003055	**
## vote_share_lastelection	0.033245	*
## served_instateleg	0.778535	
## prop_topic_Agriculture	< 2e-16	***
## prop_topic_Civil.Rights	< 2e-16	***
## prop_topic_Defense	< 2e-16	***
## prop_topic_Domestic.Commerce	< 2e-16	***
## prop_topic_Education	< 2e-16	***
## prop_topic_Energy	< 2e-16	***
## prop_topic_Environment	< 2e-16	***
## prop_topic_Foreign.Trade	< 2e-16	***
## prop_topic_Government.Operations	< 2e-16	***
## prop_topic_Health	< 2e-16	***
## prop_topic_Housing	< 2e-16	***
## prop_topic_Immigration	5.62e-05	***
## prop_topic_International.Affairs	< 2e-16	***
## prop_topic_Labor	< 2e-16	***
## prop_topic_Law.and.Crime	< 2e-16	***
## prop_topic_Macroeconomics	< 2e-16	***
## prop_topic_Public.Lands	< 2e-16	***
## prop_topic_Social.Welfare	< 2e-16	***

```
## prop_topic_Technology          < 2e-16 ***
## prop_topic_Transportation      < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 60 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)          if you need it
```

```
vif(emi_model_morecontrols1973)
```

```
##                                GVIF Df GVIF^(1/(2*Df))
## ideology                      1.207776 1          1.098989
## congress                      4.454888 25          1.030331
## chamber_opposition            1.252146 1          1.118993
## president_opposition          1.103971 1          1.050700
## republican                    1.145147 1          1.070116
## female                       1.137576 1          1.066572
## senate                       1.137373 1          1.066477
## seniority_terms               1.758314 1          1.326015
## first_term                    1.335470 1          1.155626
## committee_chair               1.175117 1          1.084028
## subcommittee_chair            1.230624 1          1.109335
## majority_party_leadership     1.387520 1          1.177930
## minority_party_leadership     1.371219 1          1.170991
## vote_share_lastelection       1.192090 1          1.091829
## served_instateleg             1.007534 1          1.003760
## prop_topic_Agriculture         1.385452 1          1.177052
## prop_topic_Civil.Rights        1.447957 1          1.203311
## prop_topic_Defense             2.201533 1          1.483756
## prop_topic_Domestic.Commerce  1.459007 1          1.207893
## prop_topic_Education           1.344254 1          1.159420
## prop_topic_Energy              1.609739 1          1.268755
## prop_topic_Environment         1.339522 1          1.157377
## prop_topic_Foreign.Trade       1.210657 1          1.100299
## prop_topic_Government.Operations 2.633113 1          1.622687
## prop_topic_Health              1.799615 1          1.341497
## prop_topic_Housing             1.125862 1          1.061066
## prop_topic_Immigration          1.119130 1          1.057890
## prop_topic_International.Affairs 1.770838 1          1.330728
## prop_topic_Labor               1.173242 1          1.083163
## prop_topic_Law.and.Crime       1.541137 1          1.241425
## prop_topic_Macroeconomics      1.961573 1          1.400562
## prop_topic_Public.Lands        1.435390 1          1.198078
## prop_topic_Social.Welfare      1.214008 1          1.101820
## prop_topic_Technology          1.237957 1          1.112635
## prop_topic_Transportation      1.329649 1          1.153104
```

```
# coef plot
```

```
vcov_full <- vcovCR(
  emi_model_morecontrols1973,
```

```

cluster = df_morecontrols1973$bioguide_id,
type = "CR2"
)

focal_terms <- c(
  "ideology",
  "vote_share_lastelection",
  "chamber_opposition",
  "female"
)

plot_df <- modelplot(
  list("Full model" = emi_model_morecontrols1973),
  vcov = list(vcov_full),
  conf_level = 0.95,
  draw = FALSE
) |>
  filter(term %in% focal_terms) |>
  mutate(
    label = dplyr::recode(
      term,
      ideology = "Ideological extremity",
      vote_share_lastelection = "Vote share",
      chamber_opposition = "Opposition",
      female = "Female"
    ),
    label = factor(
      label,
      levels = c(
        "Female",
        "Opposition",
        "Vote share",
        "Ideological extremity"
      )
    )
  )

p <- ggplot(
  plot_df,
  aes(
    x = estimate,
    y = label,
    xmin = conf.low,
    xmax = conf.high
  )
) +
  geom_vline(
    xintercept = 0,
    linetype = "dashed",
    linewidth = 0.5
  ) +
  geom_errorbarh(height = 0.12, linewidth = 0.7) +
  geom_point(size = 2.8) +

```

```

labs(
  x = "Coefficient estimate",
  y = NULL
) +
theme_minimal(base_size = 12)

ggsave(
  filename = "../output/EMI_characteristics-coeff_plot.pdf",
  plot = p,
  width = 8,
  height = 4.5,
  units = "in"
)

ggsave(
  filename = "../output/EMI_characteristics-coeff_plot.png",
  plot = p,
  width = 8,
  height = 4.5,
  units = "in",
  dpi = 300
)

modelsummary(
  list(emi_model_base, emi_model, emi_model_morecontrols1973),
  vcov = list(vcovCR(emi_model_base, cluster = df$bioguide_id, type = "CR2"),
              vcovCR(emi_model, cluster = df$bioguide_id, type = "CR2"),
              vcovCR(emi_model_morecontrols1973, cluster = df_morecontrols1973$bioguide_id, type = "CR2")),
  coef_omit = "^bioguide_id|^congress",
  gof_omit = "AIC|BIC|RMSE|Log.Lik.",
  output = "../output/EMI-characteristics-coeff_summary.html",
  fmt = NULL,
  estimate = "{f_est(estimate)} [{f_est(conf.low)}, {f_est(conf.high)}]",
  statistic = "p={f_p(p.value)}", conf_level = 0.95,
  notes = "Values in square brackets represent 95% confidence intervals based on CR2 cluster-robust standard errors"
)

tex <- modelsummary(
  list(emi_model_base, emi_model, emi_model_morecontrols1973),
  vcov = list(vcovCR(emi_model_base, cluster = df$bioguide_id, type = "CR2"),
              vcovCR(emi_model, cluster = df$bioguide_id, type = "CR2"),
              vcovCR(emi_model_morecontrols1973, cluster = df_morecontrols1973$bioguide_id, type = "CR2")),
  coef_omit = "^bioguide_id|^congress",
  gof_omit = "AIC|BIC|RMSE|Log.Lik.",
  output = "tinytable",
  fmt = NULL,
  estimate = "{f_est(estimate)}",
  statistic = c(
    "[{f_est(conf.low)}, {f_est(conf.high)}]",
    "p={f_p(p.value)}"
  )
)

```

```

),
conf_level=0.95,
notes = "Values in square brackets represent 95% confidence intervals based on cluster-robust standard errors"
)|>
theme_tt("tabular")
save_tt(tex, "../output/EMI-characteristics-meff_summary.tex", overwrite = TRUE)

```

les model

```

les_model_base <- lmer(log_les ~ 1 + ideology + congress
  + chamber_opposition + president_opposition + republican
  + female + senate
  + latino_a
  + african_american
  + seniority_terms + first_term
  + committee_chair + subcommittee_chair
  + majority_party_leadership + minority_party_leadership
  + vote_share_lastelection + served_instateleg
  + (1|bioguide_id),
  data = df_morecontrols1973)
summary(les_model_base)

```

```

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## log_les ~ 1 + ideology + congress + chamber_opposition + president_opposition +
## republican + female + senate + latino_a + african_american +
## seniority_terms + first_term + committee_chair + subcommittee_chair +
## majority_party_leadership + minority_party_leadership + vote_share_lastelection +
## served_instateleg + (1 | bioguide_id)
## Data: df_morecontrols1973
##
## REML criterion at convergence: 36817.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.0916 -0.6669 -0.1432  0.5455  6.5805
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## bioguide_id (Intercept) 0.2980      0.5459
## Residual                0.7031      0.8385
## Number of obs: 13682, groups: bioguide_id, 2454
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -9.786e-02  4.935e-02  9.063e+03  -1.983  0.047400 *
## ideology       -3.664e-02  1.470e-02  2.479e+03  -2.492  0.012780 *
## congress94      2.852e-03  5.328e-02  1.225e+04   0.054  0.957307
## congress95      3.599e-02  5.462e-02  1.279e+04   0.659  0.509918

```

```

## congress96          1.600e-02  5.504e-02  1.307e+04   0.291 0.771311
## congress97          2.689e-02  5.570e-02  1.330e+04   0.483 0.629286
## congress98         -4.860e-03  5.587e-02  1.341e+04  -0.087 0.930680
## congress99          7.964e-03  5.612e-02  1.349e+04   0.142 0.887144
## congress100         8.712e-02  5.613e-02  1.354e+04   1.552 0.120626
## congress101         1.699e-01  5.652e-02  1.359e+04   3.006 0.002651 **
## congress102         1.406e-01  5.657e-02  1.362e+04   2.484 0.012988 *
## congress103         5.726e-02  5.725e-02  1.364e+04   1.000 0.317180
## congress104         1.101e-01  5.738e-02  1.362e+04   1.918 0.055069 .
## congress105         2.258e-01  5.799e-02  1.358e+04   3.893 9.96e-05 ***
## congress106         8.728e-02  5.781e-02  1.350e+04   1.510 0.131095
## congress107         1.733e-01  5.838e-02  1.341e+04   2.969 0.002997 **
## congress108         1.422e-01  5.865e-02  1.328e+04   2.425 0.015343 *
## congress109         1.237e-01  5.896e-02  1.313e+04   2.098 0.035919 *
## congress110         1.768e-01  5.914e-02  1.291e+04   2.990 0.002795 **
## congress111         1.147e-01  5.898e-02  1.265e+04   1.944 0.051900 .
## congress112         4.886e-02  5.966e-02  1.240e+04   0.819 0.412842
## congress113         5.319e-02  5.989e-02  1.213e+04   0.888 0.374501
## congress114         6.221e-02  6.030e-02  1.181e+04   1.032 0.302276
## congress115         9.215e-02  6.061e-02  1.137e+04   1.520 0.128457
## congress116         6.786e-02  6.106e-02  1.097e+04   1.111 0.266460
## congress117         3.258e-02  6.162e-02  1.053e+04   0.529 0.596982
## congress118         7.219e-02  6.211e-02  1.017e+04   1.162 0.245170
## chamber_opposition   4.884e-02  1.682e-02  1.248e+04   2.903 0.003703 **
## president_opposition -2.239e-03  1.595e-02  1.264e+04  -0.140 0.888344
## republican          -3.262e-02  3.073e-02  2.344e+03  -1.061 0.288682
## female              1.508e-01  4.399e-02  2.299e+03   3.429 0.000617 ***
## senate             -9.850e-02  3.331e-02  5.285e+03  -2.957 0.003121 **
## latino_a            2.787e-02  7.399e-02  2.201e+03   0.377 0.706414
## african_american    -1.616e-01  6.298e-02  2.221e+03  -2.566 0.010357 *
## seniority_terms     -1.320e-01  1.194e-02  9.630e+03 -11.055 < 2e-16 ***
## first_term          -1.930e-01  2.448e-02  1.274e+04  -7.884 3.44e-15 ***
## committee_chair      1.109e-01  3.416e-02  1.299e+04   3.245 0.001175 **
## subcommittee_chair   3.343e-02  2.018e-02  1.338e+04   1.657 0.097585 .
## majority_party_leadership 2.155e-01  6.170e-02  1.277e+04   3.493 0.000479 ***
## minority_party_leadership 1.681e-02  6.210e-02  1.267e+04   0.271 0.786636
## vote_share_lastelection -2.805e-02  1.016e-02  1.348e+04  -2.761 0.005764 **
## served_instateleg    2.563e-02  2.826e-02  2.142e+03   0.907 0.364411
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

##
## Correlation matrix not shown by default, as p = 42 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)      if you need it

```

```
vif(les_model_base)
```

```

##              GVIF Df GVIF^(1/(2*Df))
## ideology          1.225879  1          1.107194
## congress           1.587934 25          1.009292
## chamber_opposition 1.247593  1          1.116957
## president_opposition 1.094402  1          1.046136

```

```
## republican          1.229245  1      1.108713
## female              1.125904  1      1.061086
## senate              1.099457  1      1.048550
## latino_a            1.046404  1      1.022939
## african_american    1.174556  1      1.083769
## seniority_terms     1.623634  1      1.274219
## first_term          1.334933  1      1.155393
## committee_chair     1.175245  1      1.084087
## subcommittee_chair  1.222766  1      1.105787
## majority_party_leadership 1.399277  1      1.182910
## minority_party_leadership 1.386104  1      1.177329
## vote_share_lastelection 1.216913  1      1.103138
## served_instateleg    1.006668  1      1.003329
```

```
cor.test(df_morecontrols1973$emi, df_morecontrols1973$log_les)
```

```
##
## Pearson's product-moment correlation
##
## data: df_morecontrols1973$emi and df_morecontrols1973$log_les
## t = 16.65, df = 13680, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.1244698 0.1573171
## sample estimates:
##      cor
## 0.1409322
```

```
les_model <- lmer(log_les ~ 1 + emi + ideology + congress
  + chamber_opposition + president_opposition + republican
  + female + senate
  + latino_a
  + african_american
  + seniority_terms + first_term
  + committee_chair + subcommittee_chair
  + majority_party_leadership + minority_party_leadership
  + vote_share_lastelection + served_instateleg
  + (1|bioguide_id),
  data = df_morecontrols1973)
summary(les_model)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: log_les ~ 1 + emi + ideology + congress + chamber_opposition +
##   president_opposition + republican + female + senate + latino_a +
##   african_american + seniority_terms + first_term + committee_chair +
##   subcommittee_chair + majority_party_leadership + minority_party_leadership +
##   vote_share_lastelection + served_instateleg + (1 | bioguide_id)
## Data: df_morecontrols1973
##
## REML criterion at convergence: 36557.7
##
## Scaled residuals:
```

```

##      Min      1Q  Median      3Q      Max
## -3.1971 -0.6645 -0.1440  0.5387  6.4946
##
## Random effects:
##      Groups      Name      Variance Std.Dev.
## bioguide_id (Intercept) 0.2876   0.5363
## Residual                0.6909   0.8312
## Number of obs: 13682, groups: bioguide_id, 2454
##
## Fixed effects:
##
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    -1.044e-01  4.880e-02  9.121e+03  -2.139  0.032491 *
## emi             1.634e-01  9.951e-03  1.283e+04  16.418  < 2e-16 ***
## ideology        2.712e-03  1.469e-02  2.583e+03   0.185  0.853540
## congress94      -1.429e-02  5.282e-02  1.226e+04  -0.271  0.786696
## congress95       2.371e-02  5.413e-02  1.279e+04   0.438  0.661316
## congress96       3.966e-02  5.455e-02  1.308e+04   0.727  0.467125
## congress97       7.868e-02  5.527e-02  1.331e+04   1.423  0.154642
## congress98       1.572e-02  5.536e-02  1.341e+04   0.284  0.776392
## congress99       4.487e-02  5.563e-02  1.349e+04   0.806  0.419989
## congress100      1.060e-01  5.561e-02  1.354e+04   1.906  0.056715 .
## congress101      1.906e-01  5.600e-02  1.359e+04   3.403  0.000668 ***
## congress102      1.577e-01  5.604e-02  1.362e+04   2.814  0.004895 **
## congress103      1.237e-01  5.684e-02  1.364e+04   2.176  0.029544 *
## congress104      1.986e-01  5.708e-02  1.362e+04   3.479  0.000504 ***
## congress105      2.589e-01  5.746e-02  1.358e+04   4.505  6.69e-06 ***
## congress106      7.367e-02  5.724e-02  1.351e+04   1.287  0.198066
## congress107      1.815e-01  5.780e-02  1.342e+04   3.140  0.001694 **
## congress108      1.428e-01  5.806e-02  1.329e+04   2.460  0.013922 *
## congress109      1.035e-01  5.837e-02  1.314e+04   1.773  0.076274 .
## congress110      1.241e-01  5.862e-02  1.293e+04   2.117  0.034301 *
## congress111      1.278e-01  5.838e-02  1.267e+04   2.189  0.028626 *
## congress112      9.051e-02  5.909e-02  1.243e+04   1.532  0.125653
## congress113      5.545e-02  5.926e-02  1.216e+04   0.936  0.349432
## congress114      3.409e-02  5.969e-02  1.185e+04   0.571  0.567887
## congress115      1.602e-02  6.014e-02  1.142e+04   0.266  0.789995
## congress116      1.311e-02  6.049e-02  1.103e+04   0.217  0.828457
## congress117      3.887e-02  6.095e-02  1.058e+04   0.638  0.523724
## congress118      8.360e-02  6.143e-02  1.023e+04   1.361  0.173610
## chamber_opposition 6.683e-02  1.671e-02  1.250e+04   4.000  6.36e-05 ***
## president_opposition 1.267e-02  1.583e-02  1.266e+04   0.800  0.423472
## republican       -2.307e-02  3.030e-02  2.344e+03  -0.761  0.446485
## female           1.413e-01  4.335e-02  2.301e+03   3.260  0.001132 **
## senate          -9.818e-02  3.290e-02  5.245e+03  -2.984  0.002857 **
## latino_a         2.491e-02  7.291e-02  2.202e+03   0.342  0.732613
## african_american -1.376e-01  6.208e-02  2.223e+03  -2.217  0.026748 *
## seniority_terms  -1.338e-01  1.181e-02  9.635e+03 -11.332  < 2e-16 ***
## first_term       -1.903e-01  2.426e-02  1.275e+04  -7.845  4.68e-15 ***
## committee_chair   4.586e-02  3.408e-02  1.304e+04   1.346  0.178461
## subcommittee_chair -1.632e-02  2.022e-02  1.342e+04  -0.807  0.419478
## majority_party_leadership 2.342e-01  6.114e-02  1.278e+04   3.831  0.000128 ***
## minority_party_leadership -7.397e-04  6.155e-02  1.269e+04  -0.012  0.990411
## vote_share_lastelection -2.837e-02  1.006e-02  1.347e+04  -2.821  0.004792 **
## served_instateleg 2.577e-02  2.784e-02  2.143e+03   0.926  0.354680

```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation matrix not shown by default, as p = 43 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)          if you need it
```

```
vif(les_model)
```

```
##
##              GVIF Df GVIF^(1/(2*Df))
## emi          1.188089 1          1.089995
## ideology     1.259359 1          1.122212
## congress     1.702269 25          1.010696
## chamber_opposition 1.252981 1          1.119366
## president_opposition 1.098097 1          1.047901
## republican   1.229860 1          1.108990
## female       1.126095 1          1.061176
## senate       1.099781 1          1.048705
## latino_a     1.046433 1          1.022953
## african_american 1.175525 1          1.084216
## seniority_terms 1.620685 1          1.273061
## first_term   1.335202 1          1.155510
## committee_chair 1.191819 1          1.091705
## subcommittee_chair 1.251405 1          1.118662
## majority_party_leadership 1.400557 1          1.183451
## minority_party_leadership 1.387363 1          1.177864
## vote_share_lastelection 1.217558 1          1.103430
## served_instateleg 1.006715 1          1.003352
```

```
les_model_more <- lmer(log_les ~ 1 + emi + ideology + congress
  + chamber_opposition + president_opposition + republican
  + female + senate
  + latino_a
  + african_american
  + seniority_terms + first_term
  + committee_chair + subcommittee_chair
  + majority_party_leadership + minority_party_leadership
  + vote_share_lastelection + served_instateleg
  + prop_topic_Agriculture + prop_topic_Civil.Rights
  + prop_topic_Defense + prop_topic_Domestic.Commerce
  + prop_topic_Education + prop_topic_Energy
  + prop_topic_Environment + prop_topic_Foreign.Trade
  + prop_topic_Government.Operations + prop_topic_Health
  + prop_topic_Housing + prop_topic_Immigration
  + prop_topic_International.Affairs + prop_topic_Labor
  + prop_topic_Law.and.Crime + prop_topic_Macroeconomics
  + prop_topic_Public.Lands + prop_topic_Social.Welfare
  + prop_topic_Technology + prop_topic_Transportation
  + log_speechcount + log_tokencount
  + (1|bioguide_id),
  data = df_morecontrols1973)
```

```
summary(les_model_more)
```

```

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: log_les ~ 1 + emi + ideology + congress + chamber_opposition +
##     president_opposition + republican + female + senate + latino_a +
##     african_american + seniority_terms + first_term + committee_chair +
##     subcommittee_chair + majority_party_leadership + minority_party_leadership +
##     vote_share_lastelection + served_instateleg + prop_topic_Agriculture +
##     prop_topic_Civil.Rights + prop_topic_Defense + prop_topic_Domestic.Commerce +
##     prop_topic_Education + prop_topic_Energy + prop_topic_Environment +
##     prop_topic_Foreign.Trade + prop_topic_Government.Operations +
##     prop_topic_Health + prop_topic_Housing + prop_topic_Immigration +
##     prop_topic_International.Affairs + prop_topic_Labor + prop_topic_Law.and.Crime +
##     prop_topic_Macroeconomics + prop_topic_Public.Lands + prop_topic_Social.Welfare +
##     prop_topic_Technology + prop_topic_Transportation + log_speechcount +
##     log_tokencount + (1 | bioguide_id)
## Data: df_morecontrols1973
##
## REML criterion at convergence: 35807
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.8079 -0.6719 -0.1374  0.5435  6.4172
##
## Random effects:
##   Groups      Name      Variance Std.Dev.
## bioguide_id (Intercept) 0.2130   0.4615
## Residual                0.6652   0.8156
## Number of obs: 13682, groups: bioguide_id, 2454
##
## Fixed effects:
##


|             | Estimate   | Std. Error | df        | t value |
|-------------|------------|------------|-----------|---------|
| (Intercept) | -2.167e+00 | 1.688e-01  | 1.083e+04 | -12.835 |
| emi         | 1.293e-01  | 1.115e-02  | 1.311e+04 | 11.595  |
| ideology    | -4.705e-02 | 1.356e-02  | 2.554e+03 | -3.469  |
| congress94  | 2.838e-03  | 5.185e-02  | 1.225e+04 | 0.055   |
| congress95  | 5.394e-03  | 5.310e-02  | 1.281e+04 | 0.102   |
| congress96  | 4.262e-02  | 5.343e-02  | 1.310e+04 | 0.798   |
| congress97  | 1.599e-01  | 5.485e-02  | 1.334e+04 | 2.916   |
| congress98  | 6.393e-02  | 5.517e-02  | 1.344e+04 | 1.159   |
| congress99  | 9.634e-02  | 5.552e-02  | 1.350e+04 | 1.735   |
| congress100 | 1.408e-01  | 5.546e-02  | 1.353e+04 | 2.539   |
| congress101 | 2.366e-01  | 5.616e-02  | 1.357e+04 | 4.213   |
| congress102 | 2.048e-01  | 5.606e-02  | 1.360e+04 | 3.653   |
| congress103 | 1.631e-01  | 5.717e-02  | 1.362e+04 | 2.853   |
| congress104 | 1.748e-01  | 5.696e-02  | 1.361e+04 | 3.069   |
| congress105 | 3.269e-01  | 5.720e-02  | 1.358e+04 | 5.715   |
| congress106 | 1.266e-01  | 5.776e-02  | 1.354e+04 | 2.192   |
| congress107 | 3.497e-01  | 5.810e-02  | 1.348e+04 | 6.019   |
| congress108 | 2.810e-01  | 5.875e-02  | 1.338e+04 | 4.783   |
| congress109 | 2.467e-01  | 5.875e-02  | 1.327e+04 | 4.199   |
| congress110 | 2.551e-01  | 5.878e-02  | 1.311e+04 | 4.340   |
| congress111 | 3.515e-01  | 6.043e-02  | 1.301e+04 | 5.817   |
| congress112 | 3.316e-01  | 6.041e-02  | 1.286e+04 | 5.489   |
| congress113 | 3.099e-01  | 6.002e-02  | 1.258e+04 | 5.163   |


```

## congress114	2.582e-01	6.092e-02	1.234e+04	4.239
## congress115	2.960e-01	6.204e-02	1.208e+04	4.771
## congress116	3.490e-01	6.219e-02	1.174e+04	5.612
## congress117	4.631e-01	6.343e-02	1.141e+04	7.301
## congress118	4.527e-01	6.254e-02	1.104e+04	7.237
## chamber_opposition	3.570e-02	1.641e-02	1.260e+04	2.176
## president_opposition	3.956e-03	1.553e-02	1.279e+04	0.255
## republican	-3.073e-02	2.757e-02	2.224e+03	-1.115
## female	4.927e-02	3.988e-02	2.290e+03	1.235
## senate	-4.770e-01	3.752e-02	6.016e+03	-12.713
## latino_a	-1.360e-03	6.665e-02	2.153e+03	-0.020
## african_american	-9.852e-02	5.728e-02	2.240e+03	-1.720
## seniority_terms	-1.681e-01	1.151e-02	9.818e+03	-14.605
## first_term	-1.566e-01	2.374e-02	1.282e+04	-6.594
## committee_chair	-1.400e-02	3.345e-02	1.323e+04	-0.418
## subcommittee_chair	-3.822e-02	1.973e-02	1.352e+04	-1.937
## majority_party_leadership	2.390e-01	5.970e-02	1.295e+04	4.003
## minority_party_leadership	-3.627e-02	6.011e-02	1.283e+04	-0.603
## vote_share_lastelection	-2.416e-02	9.713e-03	1.313e+04	-2.487
## served_instateleg	2.202e-02	2.515e-02	2.031e+03	0.876
## prop_topic_Agriculture	2.321e-03	1.101e-02	1.195e+04	0.211
## prop_topic_Civil.Rights	7.946e-03	1.061e-02	1.340e+04	0.749
## prop_topic_Defense	-1.047e-02	1.447e-02	1.309e+04	-0.724
## prop_topic_Domestic.Commerce	2.073e-02	1.047e-02	1.341e+04	1.980
## prop_topic_Education	-1.522e-02	1.013e-02	1.351e+04	-1.502
## prop_topic_Energy	5.654e-04	1.109e-02	1.341e+04	0.051
## prop_topic_Environment	1.526e-02	1.013e-02	1.351e+04	1.506
## prop_topic_Foreign.Trade	1.176e-02	8.975e-03	1.362e+04	1.310
## prop_topic_Government.Operations	1.027e-03	1.451e-02	1.359e+04	0.071
## prop_topic_Health	3.338e-02	1.225e-02	1.325e+04	2.725
## prop_topic_Housing	-9.766e-03	8.970e-03	1.333e+04	-1.089
## prop_topic_Immigration	5.958e-03	8.930e-03	1.354e+04	0.667
## prop_topic_International.Affairs	5.213e-03	1.269e-02	1.237e+04	0.411
## prop_topic_Labor	1.285e-02	9.025e-03	1.354e+04	1.424
## prop_topic_Law.and.Crime	5.353e-02	1.056e-02	1.340e+04	5.070
## prop_topic_Macroeconomics	-2.590e-02	1.183e-02	1.344e+04	-2.189
## prop_topic_Public.Lands	1.204e-01	1.151e-02	1.269e+04	10.461
## prop_topic_Social.Welfare	1.938e-02	9.024e-03	1.361e+04	2.148
## prop_topic_Technology	-1.529e-03	9.800e-03	1.353e+04	-0.156
## prop_topic_Transportation	3.014e-02	1.056e-02	1.307e+04	2.855
## log_speechcount	1.770e-01	2.791e-02	1.190e+04	6.343
## log_tokencount	1.281e-01	2.666e-02	1.148e+04	4.803
##	Pr(> t)			
## (Intercept)	< 2e-16 ***			
## emi	< 2e-16 ***			
## ideology	0.000531 ***			
## congress94	0.956347			
## congress95	0.919102			
## congress96	0.425095			
## congress97	0.003552 **			
## congress98	0.246537			
## congress99	0.082701 .			
## congress100	0.011116 *			
## congress101	2.54e-05 ***			

## congress102	0.000260	***
## congress103	0.004336	**
## congress104	0.002153	**
## congress105	1.12e-08	***
## congress106	0.028392	*
## congress107	1.80e-09	***
## congress108	1.74e-06	***
## congress109	2.70e-05	***
## congress110	1.43e-05	***
## congress111	6.13e-09	***
## congress112	4.12e-08	***
## congress113	2.46e-07	***
## congress114	2.26e-05	***
## congress115	1.86e-06	***
## congress116	2.04e-08	***
## congress117	3.05e-13	***
## congress118	4.87e-13	***
## chamber_opposition	0.029584	*
## president_opposition	0.798939	
## republican	0.265137	
## female	0.216819	
## senate	< 2e-16	***
## latino_a	0.983728	
## african_american	0.085578	.
## seniority_terms	< 2e-16	***
## first_term	4.45e-11	***
## committee_chair	0.675644	
## subcommittee_chair	0.052739	.
## majority_party_leadership	6.29e-05	***
## minority_party_leadership	0.546228	
## vote_share_lastelection	0.012883	*
## served_instateleg	0.381307	
## prop_topic_Agriculture	0.833068	
## prop_topic_Civil.Rights	0.453854	
## prop_topic_Defense	0.469046	
## prop_topic_Domestic.Commerce	0.047677	*
## prop_topic_Education	0.133009	
## prop_topic_Energy	0.959329	
## prop_topic_Environment	0.132028	
## prop_topic_Foreign.Trade	0.190189	
## prop_topic_Government.Operations	0.943577	
## prop_topic_Health	0.006443	**
## prop_topic_Housing	0.276272	
## prop_topic_Immigration	0.504693	
## prop_topic_International.Affairs	0.681192	
## prop_topic_Labor	0.154467	
## prop_topic_Law.and.Crime	4.03e-07	***
## prop_topic_Macroeconomics	0.028613	*
## prop_topic_Public.Lands	< 2e-16	***
## prop_topic_Social.Welfare	0.031726	*
## prop_topic_Technology	0.876012	
## prop_topic_Transportation	0.004311	**
## log_speechcount	2.34e-10	***
## log_tokencount	1.58e-06	***

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 65 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)          if you need it
```

```
vif(les_model_more)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## emi          1.622588 1      1.273808
## ideology     1.304722 1      1.142244
## congress     5.496126 25      1.034668
## chamber_opposition 1.268176 1      1.126133
## president_opposition 1.111050 1      1.054064
## republican   1.245172 1      1.115873
## female       1.164221 1      1.078991
## senate       1.662681 1      1.289450
## latino_a     1.072940 1      1.035828
## african_american 1.231258 1      1.109621
## seniority_terms 1.668994 1      1.291895
## first_term   1.347118 1      1.160654
## committee_chair 1.216212 1      1.102820
## subcommittee_chair 1.267903 1      1.126012
## majority_party_leadership 1.417266 1      1.190490
## minority_party_leadership 1.403751 1      1.184800
## vote_share_lastelection 1.233283 1      1.110533
## served_instateg 1.010428 1      1.005200
## prop_topic_Agriculture 1.460125 1      1.208356
## prop_topic_Civil.Rights 1.521637 1      1.233547
## prop_topic_Defense 2.534649 1      1.592058
## prop_topic_Domestic.Commerce 1.552229 1      1.245885
## prop_topic_Education 1.465785 1      1.210696
## prop_topic_Energy 1.744947 1      1.320964
## prop_topic_Environment 1.450531 1      1.204380
## prop_topic_Foreign.Trade 1.233108 1      1.110454
## prop_topic_Government.Operations 2.842938 1      1.686101
## prop_topic_Health 1.978292 1      1.406518
## prop_topic_Housing 1.147301 1      1.071121
## prop_topic_Immigration 1.147974 1      1.071436
## prop_topic_International.Affairs 1.871049 1      1.367863
## prop_topic_Labor 1.200003 1      1.095447
## prop_topic_Law.and.Crime 1.598343 1      1.264256
## prop_topic_Macroeconomics 2.002234 1      1.415003
## prop_topic_Public.Lands 1.630367 1      1.276858
## prop_topic_Social.Welfare 1.247044 1      1.116711
## prop_topic_Technology 1.362027 1      1.167059
## prop_topic_Transportation 1.442362 1      1.200984
## log_speechcount 9.208246 1      3.034509
## log_tokencount 10.188904 1      3.192006
```

```

# coef plot

vcov_full <- vcovCR(
  les_model_more,
  cluster = df_morecontrols1973$bioguide_id,
  type = "CR2"
)

focal_terms <- c(
  "emi",
  "ideology",
  "vote_share_lastelection",
  "chamber_opposition",
  "female"
)

plot_df <- modelplot(
  list("Full model" = les_model_more),
  vcov = list(vcov_full),
  conf_level = 0.95,
  draw = FALSE
) |>
  filter(term %in% focal_terms) |>
  mutate(
    label = dplyr::recode(
      term,
      emi = "EMI",
      ideology = "Ideological extremity",
      vote_share_lastelection = "Vote share",
      chamber_opposition = "Opposition",
      female = "Female"
    ),
    label = factor(
      label,
      levels = c(
        "Female",
        "Opposition",
        "Vote share",
        "Ideological extremity",
        "EMI"
      )
    )
  )

p <- ggplot(
  plot_df,
  aes(
    x = estimate,
    y = label,
    xmin = conf.low,
    xmax = conf.high
  )
) +

```

```

geom_vline(
  xintercept = 0,
  linetype = "dashed",
  linewidth = 0.5
) +
geom_errorbarh(height = 0.12, linewidth = 0.7) +
geom_point(size = 2.8) +
labs(
  x = "Coefficient estimate",
  y = NULL
) +
theme_minimal(base_size = 12)

ggsave(
  filename = "../output/EMI_LES-coeff_plot.pdf",
  plot = p,
  width = 8,
  height = 4.5,
  units = "in"
)

ggsave(
  filename = "../output/EMI_LES-coeff_plot.png",
  plot = p,
  width = 8,
  height = 4.5,
  units = "in",
  dpi = 300
)

modelsummary(
  list(les_model_base, les_model, les_model_more),
  vcov = list(vcovCR(les_model_base, cluster = df_morecontrols1973$bioguide_id, type = "CR2"),
              vcovCR(les_model, cluster = df_morecontrols1973$bioguide_id, type = "CR2"),
              vcovCR(les_model_more, cluster = df_morecontrols1973$bioguide_id, type = "CR2")),
  coef_omit = "^bioguide_id|^congress",
  gof_omit = "AIC|BIC|RMSE|Log.Lik.",
  output = "../output/LES-EMI-meff_summary.html",
  fmt = NULL,
  estimate = "{f_est(estimate)} [{f_est(conf.low)}, {f_est(conf.high)}]",
  statistic = "p={f_p(p.value)}",
  conf_level = 0.95,
  notes = "Values in square brackets represent 95% confidence intervals based on cluster-robust standard errors"
)

tex <- modelsummary(
  list(les_model_base, les_model, les_model_more),
  vcov = list(vcovCR(les_model_base, cluster = df_morecontrols1973$bioguide_id, type = "CR2"),
              vcovCR(les_model, cluster = df_morecontrols1973$bioguide_id, type = "CR2"),
              vcovCR(les_model_more, cluster = df_morecontrols1973$bioguide_id, type = "CR2")),
  coef_omit = "^bioguide_id|^congress",
  gof_omit = "AIC|BIC|RMSE|Log.Lik.",
  output = "../output/LES-EMI-meff_summary.tex",
  fmt = NULL,
  estimate = "{f_est(estimate)} [{f_est(conf.low)}, {f_est(conf.high)}]",
  statistic = "p={f_p(p.value)}",
  conf_level = 0.95,
  notes = "Values in square brackets represent 95% confidence intervals based on cluster-robust standard errors"
)

```

```

coef_omit = "^bioguide_id|^congress",
gof_omit="AIC|BIC|RMSE|Log.Lik.",
output = "tinytable",
fmt = NULL,
estimate = "{f_est(estimate)}",
statistic = c(
  "[f_est(conf.low)}, {f_est(conf.high)}]",
  "p={f_p(p.value)}"
),
conf_level=0.95,
notes = "Values in square brackets represent 95% confidence intervals based on cluster-robust standard errors"
)|>
  theme_tt("tabular")
  save_tt(tex, "../output/LES-EMI-meff_summary.tex", overwrite = TRUE)

```

Twitter-Congress model

```

df_tw_congress <- read.csv("../data/twitter_speech_sbert_llms.csv")
df_tw_congress <- subset(
  df_tw_congress,
  party_name %in% c("Democratic Party", "Republican Party")
)
df_tw_congress$emi <- df_tw_congress$emi_mean
df_tw_congress$emi_sc <- scale(df_tw_congress$emi)
df_tw_congress$bioguide_id <- factor(df_tw_congress$bioguide_id)
df_tw_congress$year_quarter <- factor(df_tw_congress$year_quarter)

df_tw_congress$republican <- ifelse(
df_tw_congress$party_name == "Republican Party",
  1,
  0
)

df_tw_congress$senate <- ifelse(
df_tw_congress$chamber == "S",
  1,
  0
)

tw <- subset(df_tw_congress, channel == "Twitter")
floor <- subset(df_tw_congress, channel == "Floor")
merged <- merge(
  tw,
  floor,
  by = c("bioguide_id", "year_quarter", "chamber"),
  suffixes = c("_tw", "_floor")
)

cor.test(
  merged$emi_mean_tw,

```

```
merged$emi_mean_floor
)
```

```
##
## Pearson's product-moment correlation
##
## data: merged$emi_mean_tw and merged$emi_mean_floor
## t = 31.911, df = 4189, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.4175280 0.4662491
## sample estimates:
## cor
## 0.4422147
```

```
merged$emi_mean_tw <- scale(merged$emi_mean_tw)
merged$emi_mean_floor <- scale(merged$emi_mean_floor)
```

```
# Twitter as predictor for floor
model_floor_tw_base <- lmer(
  emi_mean_floor ~ emi_mean_tw + year_quarter +
    (1 | bioguide_id),
  data = merged
)
summary(model_floor_tw_base)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: emi_mean_floor ~ emi_mean_tw + year_quarter + (1 | bioguide_id)
## Data: merged
##
## REML criterion at convergence: 9245
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -4.6138 -0.6083 -0.0037 0.5883 4.7823
##
## Random effects:
## Groups Name Variance Std.Dev.
## bioguide_id (Intercept) 0.3719 0.6099
## Residual 0.4235 0.6508
## Number of obs: 4191, groups: bioguide_id, 467
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) -0.15402 0.08090 3986.03508 -1.904 0.056992 .
## emi_mean_tw 0.36172 0.01456 4122.64322 24.848 < 2e-16 ***
## year_quarter2013Q2 -0.05398 0.09212 3790.32841 -0.586 0.557947
## year_quarter2013Q3 -0.04801 0.09799 3750.77491 -0.490 0.624226
## year_quarter2013Q4 -0.05527 0.09891 3748.49188 -0.559 0.576348
## year_quarter2014Q1 0.07032 0.10088 3763.05962 0.697 0.485843
## year_quarter2014Q2 0.31152 0.09574 3751.02506 3.254 0.001148 **
## year_quarter2014Q3 0.15533 0.09955 3753.03836 1.560 0.118764
```

```
## year_quarter2014Q4    0.04901    0.12187 3736.28114    0.402 0.687586
## year_quarter2015Q1    0.03553    0.09651 3744.66826    0.368 0.712785
## year_quarter2015Q2    0.26940    0.09233 3790.66106    2.918 0.003546 **
## year_quarter2015Q3    0.15654    0.10003 3749.38476    1.565 0.117685
## year_quarter2015Q4    0.06476    0.09399 3756.45965    0.689 0.490895
## year_quarter2016Q1    0.12704    0.09922 3775.92679    1.280 0.200464
## year_quarter2016Q2    0.11559    0.09237 3753.78591    1.251 0.210886
## year_quarter2016Q3    0.23337    0.09874 3789.39904    2.363 0.018154 *
## year_quarter2016Q4    0.60798    0.14783 3720.20627    4.113 3.99e-05 ***
## year_quarter2017Q1    0.03529    0.09195 3780.57405    0.384 0.701140
## year_quarter2017Q2    0.28366    0.10112 3754.92989    2.805 0.005054 **
## year_quarter2017Q3    0.60346    0.09487 3785.13747    6.361 2.24e-10 ***
## year_quarter2017Q4    0.17794    0.09262 3780.15939    1.921 0.054792 .
## year_quarter2018Q1    0.17430    0.09831 3759.14845    1.773 0.076329 .
## year_quarter2018Q2    0.31908    0.09291 3802.53847    3.434 0.000601 ***
## year_quarter2018Q3    0.52904    0.10573 3754.55036    5.004 5.88e-07 ***
## year_quarter2018Q4    0.20267    0.11950 3726.92778    1.696 0.089976 .
## year_quarter2019Q1    0.12741    0.09556 3782.15758    1.333 0.182497
## year_quarter2019Q2    0.38377    0.09506 3830.75604    4.037 5.52e-05 ***
## year_quarter2019Q3    0.57041    0.10944 3827.69320    5.212 1.97e-07 ***
## year_quarter2019Q4    0.38857    0.09366 3808.85105    4.149 3.41e-05 ***
## year_quarter2020Q1    0.26591    0.10050 3807.90619    2.646 0.008183 **
## year_quarter2020Q2    0.20270    0.11997 3769.81075    1.690 0.091177 .
## year_quarter2020Q3    0.21725    0.10078 3816.26435    2.156 0.031175 *
## year_quarter2020Q4    0.37318    0.10748 3793.36218    3.472 0.000522 ***
## year_quarter2021Q1    -0.13943    0.10083 3802.15484   -1.383 0.166801
## year_quarter2021Q2    0.41316    0.09802 3862.37128    4.215 2.55e-05 ***
## year_quarter2021Q3    0.02112    0.10921 3827.62086    0.193 0.846674
## year_quarter2021Q4    0.12262    0.10135 3818.17592    1.210 0.226400
## year_quarter2022Q1    0.06007    0.10105 3841.75245    0.594 0.552263
## year_quarter2022Q2    0.42186    0.09860 3854.92229    4.279 1.93e-05 ***
## year_quarter2022Q3    0.26820    0.10174 3802.61215    2.636 0.008418 **
## year_quarter2022Q4    0.55366    0.12398 3818.56708    4.466 8.22e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation matrix not shown by default, as p = 41 > 12.
## Use print(x, correlation=TRUE) or
##     vcov(x)           if you need it
```

```
model_floor_tw <- lmer(emi_mean_floor ~ 1 + emi_mean_tw + legislative_opposition_tw
+ presidential_opposition_tw
+ year_quarter + senate_tw
+ republican_tw + (1|bioguide_id),
  data=merged
)
summary(model_floor_tw)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: emi_mean_floor ~ 1 + emi_mean_tw + legislative_opposition_tw +
##     presidential_opposition_tw + year_quarter + senate_tw + republican_tw +
```

```

##      (1 | bioguide_id)
##      Data: merged
##
## REML criterion at convergence: 9100.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.6515 -0.6089  0.0041  0.5895  4.5971
##
## Random effects:
##      Groups      Name      Variance Std.Dev.
## bioguide_id (Intercept) 0.3573   0.5977
## Residual              0.4077   0.6385
## Number of obs: 4191, groups: bioguide_id, 467
##
## Fixed effects:
##
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      2.708e-01  8.902e-02  3.169e+03   3.042 0.002370
## emi_mean_tw       2.882e-01  1.640e-02  4.053e+03  17.575 < 2e-16
## legislative_opposition_tw -2.814e-01  2.571e-02  4.078e+03 -10.941 < 2e-16
## presidential_opposition_tw -9.140e-02  2.382e-02  4.029e+03  -3.836 0.000127
## year_quarter2013Q2      -8.392e-02  9.044e-02  3.785e+03  -0.928 0.353555
## year_quarter2013Q3      -6.922e-02  9.617e-02  3.745e+03  -0.720 0.471723
## year_quarter2013Q4      -9.126e-02  9.709e-02  3.743e+03  -0.940 0.347258
## year_quarter2014Q1       5.360e-02  9.900e-02  3.757e+03   0.541 0.588245
## year_quarter2014Q2       2.971e-01  9.397e-02  3.746e+03   3.162 0.001581
## year_quarter2014Q3       1.533e-01  9.772e-02  3.748e+03   1.569 0.116725
## year_quarter2014Q4       4.263e-02  1.196e-01  3.730e+03   0.356 0.721521
## year_quarter2015Q1       1.381e-02  9.471e-02  3.739e+03   0.146 0.884088
## year_quarter2015Q2       2.526e-01  9.061e-02  3.783e+03   2.788 0.005338
## year_quarter2015Q3       1.367e-01  9.816e-02  3.743e+03   1.393 0.163746
## year_quarter2015Q4       5.969e-02  9.227e-02  3.751e+03   0.647 0.517729
## year_quarter2016Q1       1.384e-01  9.741e-02  3.771e+03   1.421 0.155427
## year_quarter2016Q2       1.300e-01  9.070e-02  3.751e+03   1.434 0.151752
## year_quarter2016Q3       2.462e-01  9.695e-02  3.785e+03   2.540 0.011135
## year_quarter2016Q4       6.162e-01  1.451e-01  3.715e+03   4.247 2.22e-05
## year_quarter2017Q1       1.998e-02  9.028e-02  3.774e+03   0.221 0.824820
## year_quarter2017Q2       2.922e-01  9.928e-02  3.751e+03   2.943 0.003271
## year_quarter2017Q3       5.918e-01  9.315e-02  3.778e+03   6.353 2.36e-10
## year_quarter2017Q4       1.619e-01  9.093e-02  3.776e+03   1.780 0.075103
## year_quarter2018Q1       1.663e-01  9.650e-02  3.754e+03   1.723 0.084949
## year_quarter2018Q2       3.163e-01  9.123e-02  3.796e+03   3.467 0.000533
## year_quarter2018Q3       5.222e-01  1.038e-01  3.750e+03   5.032 5.09e-07
## year_quarter2018Q4       2.001e-01  1.173e-01  3.721e+03   1.706 0.088163
## year_quarter2019Q1       1.070e-01  9.384e-02  3.777e+03   1.140 0.254158
## year_quarter2019Q2       3.758e-01  9.344e-02  3.823e+03   4.022 5.89e-05
## year_quarter2019Q3       5.530e-01  1.075e-01  3.823e+03   5.143 2.84e-07
## year_quarter2019Q4       3.686e-01  9.196e-02  3.803e+03   4.008 6.24e-05
## year_quarter2020Q1       2.661e-01  9.872e-02  3.804e+03   2.696 0.007051
## year_quarter2020Q2       2.141e-01  1.178e-01  3.767e+03   1.818 0.069088
## year_quarter2020Q3       1.978e-01  9.894e-02  3.812e+03   1.999 0.045701
## year_quarter2020Q4       3.662e-01  1.055e-01  3.790e+03   3.472 0.000523
## year_quarter2021Q1      -1.809e-01  9.901e-02  3.797e+03  -1.827 0.067723
## year_quarter2021Q2       4.022e-01  9.621e-02  3.856e+03   4.181 2.97e-05

```

```

## year_quarter2021Q3      -8.188e-03  1.072e-01  3.824e+03  -0.076  0.939127
## year_quarter2021Q4      8.388e-02  9.952e-02  3.814e+03   0.843  0.399369
## year_quarter2022Q1      4.461e-02  9.918e-02  3.837e+03   0.450  0.652924
## year_quarter2022Q2      3.747e-01  9.683e-02  3.848e+03   3.870  0.000111
## year_quarter2022Q3      2.355e-01  9.987e-02  3.796e+03   2.358  0.018436
## year_quarter2022Q4      5.445e-01  1.217e-01  3.812e+03   4.474  7.90e-06
## senate_tw               -3.295e-01  6.540e-02  6.798e+02  -5.038  6.03e-07
## republican_tw           -2.786e-01  6.325e-02  4.263e+02  -4.405  1.34e-05
##
## (Intercept)             **
## emi_mean_tw              ***
## legislative_opposition_tw ***
## presidential_opposition_tw ***
## year_quarter2013Q2
## year_quarter2013Q3
## year_quarter2013Q4
## year_quarter2014Q1
## year_quarter2014Q2      **
## year_quarter2014Q3
## year_quarter2014Q4
## year_quarter2015Q1
## year_quarter2015Q2      **
## year_quarter2015Q3
## year_quarter2015Q4
## year_quarter2016Q1
## year_quarter2016Q2
## year_quarter2016Q3      *
## year_quarter2016Q4      ***
## year_quarter2017Q1
## year_quarter2017Q2      **
## year_quarter2017Q3      ***
## year_quarter2017Q4      .
## year_quarter2018Q1      .
## year_quarter2018Q2      ***
## year_quarter2018Q3      ***
## year_quarter2018Q4      .
## year_quarter2019Q1
## year_quarter2019Q2      ***
## year_quarter2019Q3      ***
## year_quarter2019Q4      ***
## year_quarter2020Q1      **
## year_quarter2020Q2      .
## year_quarter2020Q3      *
## year_quarter2020Q4      ***
## year_quarter2021Q1      .
## year_quarter2021Q2      ***
## year_quarter2021Q3
## year_quarter2021Q4
## year_quarter2022Q1
## year_quarter2022Q2      ***
## year_quarter2022Q3      *
## year_quarter2022Q4      ***
## senate_tw               ***
## republican_tw           ***

```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation matrix not shown by default, as p = 45 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)          if you need it
```

```
vif(model_floor_tw)
```

```
##                                GVIF Df GVIF^(1/(2*Df))
## emi_mean_tw                   1.417096 1          1.190418
## legislative_opposition_tw     1.114470 1          1.055685
## presidential_opposition_tw   1.234114 1          1.110907
## year_quarter                  1.128720 39          1.001554
## senate_tw                     1.011310 1          1.005639
## republican_tw                 1.015613 1          1.007776
```

```
# floor as predictor for twitter
model_tw_floor_base <- lmer(
  emi_mean_tw ~ emi_mean_floor + year_quarter +
    (1 | bioguide_id),
  data = merged
)
summary(model_tw_floor_base)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: emi_mean_tw ~ emi_mean_floor + year_quarter + (1 | bioguide_id)
##      Data: merged
##
## REML criterion at convergence: 9201.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.3162 -0.6000  0.0060  0.6134  3.5884
##
## Random effects:
##      Groups      Name      Variance Std.Dev.
## bioguide_id (Intercept) 0.3961   0.6293
## Residual                0.4162   0.6451
## Number of obs: 4191, groups: bioguide_id, 467
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   9.860e-02  8.067e-02 3.949e+03  1.222 0.221639
## emi_mean_floor 3.566e-01  1.442e-02 4.144e+03 24.736 < 2e-16 ***
## year_quarter2013Q2 7.389e-03  9.137e-02 3.790e+03  0.081 0.935552
## year_quarter2013Q3 -3.841e-02  9.717e-02 3.751e+03 -0.395 0.692614
## year_quarter2013Q4 -1.180e-01  9.806e-02 3.748e+03 -1.204 0.228762
## year_quarter2014Q1 4.456e-02  1.000e-01 3.763e+03  0.445 0.656052
## year_quarter2014Q2 -2.147e-02  9.505e-02 3.753e+03 -0.226 0.821300
```

```

## year_quarter2014Q3  5.580e-02  9.874e-02  3.753e+03  0.565 0.572034
## year_quarter2014Q4  5.560e-02  1.208e-01  3.737e+03  0.460 0.645489
## year_quarter2015Q1 -9.514e-02  9.569e-02  3.745e+03 -0.994 0.320171
## year_quarter2015Q2 -9.751e-02  9.165e-02  3.790e+03 -1.064 0.287417
## year_quarter2015Q3 -1.625e-01  9.919e-02  3.750e+03 -1.638 0.101454
## year_quarter2015Q4  8.609e-02  9.320e-02  3.757e+03  0.924 0.355708
## year_quarter2016Q1  3.123e-01  9.829e-02  3.774e+03  3.177 0.001498 **
## year_quarter2016Q2  2.168e-01  9.156e-02  3.755e+03  2.368 0.017949 *
## year_quarter2016Q3  1.842e-01  9.795e-02  3.791e+03  1.881 0.060105 .
## year_quarter2016Q4 -3.369e-02  1.469e-01  3.725e+03 -0.229 0.818558
## year_quarter2017Q1 -3.710e-01  9.101e-02  3.776e+03 -4.077 4.66e-05 ***
## year_quarter2017Q2 -2.382e-01  1.003e-01  3.754e+03 -2.374 0.017626 *
## year_quarter2017Q3 -4.057e-01  9.434e-02  3.786e+03 -4.300 1.75e-05 ***
## year_quarter2017Q4 -2.933e-01  9.179e-02  3.777e+03 -3.195 0.001410 **
## year_quarter2018Q1 -8.272e-02  9.752e-02  3.759e+03 -0.848 0.396361
## year_quarter2018Q2 -5.392e-02  9.228e-02  3.802e+03 -0.584 0.559030
## year_quarter2018Q3 -2.694e-01  1.051e-01  3.757e+03 -2.564 0.010394 *
## year_quarter2018Q4 -1.401e-01  1.185e-01  3.729e+03 -1.182 0.237183
## year_quarter2019Q1 -1.141e-01  9.477e-02  3.781e+03 -1.204 0.228729
## year_quarter2019Q2  6.623e-02  9.448e-02  3.827e+03  0.701 0.483362
## year_quarter2019Q3 -4.622e-02  1.089e-01  3.830e+03 -0.424 0.671329
## year_quarter2019Q4 -1.595e-01  9.305e-02  3.807e+03 -1.714 0.086577 .
## year_quarter2020Q1  1.716e-01  9.973e-02  3.805e+03  1.721 0.085370 .
## year_quarter2020Q2  1.992e-01  1.190e-01  3.770e+03  1.674 0.094123 .
## year_quarter2020Q3 -6.227e-02  1.000e-01  3.813e+03 -0.623 0.533555
## year_quarter2020Q4 -1.648e-01  1.067e-01  3.793e+03 -1.544 0.122557
## year_quarter2021Q1 -1.945e-01  9.998e-02  3.800e+03 -1.945 0.051859 .
## year_quarter2021Q2 -1.536e-01  9.742e-02  3.863e+03 -1.577 0.114906
## year_quarter2021Q3 -1.569e-01  1.083e-01  3.825e+03 -1.449 0.147450
## year_quarter2021Q4 -3.530e-01  1.004e-01  3.819e+03 -3.516 0.000443 ***
## year_quarter2022Q1 -1.721e-01  1.002e-01  3.840e+03 -1.717 0.086086 .
## year_quarter2022Q2 -3.488e-01  9.788e-02  3.859e+03 -3.563 0.000371 ***
## year_quarter2022Q3 -2.609e-01  1.009e-01  3.801e+03 -2.586 0.009755 **
## year_quarter2022Q4 -4.243e-01  1.231e-01  3.825e+03 -3.446 0.000574 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

##
## Correlation matrix not shown by default, as p = 41 > 12.
## Use print(x, correlation=TRUE) or
##     vcov(x)           if you need it

```

```

model_tw_floor <- lmer(emi_mean_tw ~ 1 + emi_mean_floor + legislative_opposition_tw
+ presidential_opposition_tw
+ year_quarter + senate_tw
+ republican_tw + (1|bioguide_id),
data=merged
)
summary(model_tw_floor)

```

```

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: emi_mean_tw ~ 1 + emi_mean_floor + legislative_opposition_tw +

```

```

##      presidential_opposition_tw + year_quarter + senate_tw + republican_tw +
##      (1 | bioguide_id)
##      Data: merged
##
## REML criterion at convergence: 8337.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.8256 -0.5752  0.0031  0.5977  3.8468
##
## Random effects:
##      Groups          Name          Variance Std.Dev.
## bioguide_id (Intercept) 0.3964    0.6296
## Residual                0.3297    0.5742
## Number of obs: 4191, groups: bioguide_id, 467
##
## Fixed effects:
##
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      5.661e-01  8.283e-02  2.785e+03   6.835 1.01e-11
## emi_mean_floor      2.325e-01  1.367e-02  4.144e+03  17.004 < 2e-16
## legislative_opposition_tw -3.695e-01  2.295e-02  4.025e+03 -16.100 < 2e-16
## presidential_opposition_tw -5.403e-01  1.991e-02  3.894e+03 -27.131 < 2e-16
## year_quarter2013Q2      -8.041e-03  8.149e-02  3.767e+03  -0.099 0.921400
## year_quarter2013Q3      -3.165e-02  8.659e-02  3.731e+03  -0.366 0.714757
## year_quarter2013Q4     -1.589e-01  8.737e-02  3.727e+03  -1.819 0.069028
## year_quarter2014Q1       3.843e-02  8.915e-02  3.743e+03   0.431 0.666471
## year_quarter2014Q2       2.382e-02  8.471e-02  3.733e+03   0.281 0.778527
## year_quarter2014Q3       1.029e-01  8.800e-02  3.734e+03   1.169 0.242328
## year_quarter2014Q4       5.970e-02  1.076e-01  3.719e+03   0.555 0.579162
## year_quarter2015Q1     -1.059e-01  8.524e-02  3.725e+03  -1.243 0.214102
## year_quarter2015Q2     -6.472e-02  8.169e-02  3.765e+03  -0.792 0.428270
## year_quarter2015Q3     -1.453e-01  8.836e-02  3.729e+03  -1.644 0.100179
## year_quarter2015Q4       1.161e-01  8.306e-02  3.735e+03   1.398 0.162192
## year_quarter2016Q1       3.372e-01  8.760e-02  3.753e+03   3.850 0.000120
## year_quarter2016Q2       2.614e-01  8.159e-02  3.735e+03   3.203 0.001369
## year_quarter2016Q3       2.472e-01  8.733e-02  3.769e+03   2.830 0.004673
## year_quarter2016Q4       8.655e-02  1.309e-01  3.708e+03   0.661 0.508394
## year_quarter2017Q1     -2.662e-01  8.121e-02  3.752e+03  -3.277 0.001057
## year_quarter2017Q2     -1.125e-01  8.947e-02  3.737e+03  -1.257 0.208662
## year_quarter2017Q3     -2.501e-01  8.424e-02  3.765e+03  -2.969 0.003007
## year_quarter2017Q4     -2.154e-01  8.186e-02  3.757e+03  -2.631 0.008552
## year_quarter2018Q1     -3.273e-02  8.692e-02  3.738e+03  -0.377 0.706523
## year_quarter2018Q2       3.653e-02  8.232e-02  3.777e+03   0.444 0.657222
## year_quarter2018Q3     -1.529e-01  9.371e-02  3.738e+03  -1.632 0.102826
## year_quarter2018Q4     -8.379e-02  1.056e-01  3.711e+03  -0.793 0.427541
## year_quarter2019Q1     -5.577e-02  8.454e-02  3.758e+03  -0.660 0.509442
## year_quarter2019Q2       1.706e-01  8.436e-02  3.801e+03   2.022 0.043280
## year_quarter2019Q3       6.220e-02  9.724e-02  3.809e+03   0.640 0.522467
## year_quarter2019Q4     -8.636e-02  8.303e-02  3.786e+03  -1.040 0.298344
## year_quarter2020Q1       2.332e-01  8.897e-02  3.783e+03   2.621 0.008795
## year_quarter2020Q2       1.891e-01  1.061e-01  3.750e+03   1.783 0.074709
## year_quarter2020Q3     -3.044e-02  8.922e-02  3.788e+03  -0.341 0.732950
## year_quarter2020Q4     -1.461e-01  9.514e-02  3.771e+03  -1.535 0.124802
## year_quarter2021Q1     -2.391e-01  8.917e-02  3.775e+03  -2.681 0.007373

```

```

## year_quarter2021Q2      -9.032e-02  8.696e-02  3.836e+03  -1.039  0.299041
## year_quarter2021Q3      -1.764e-01  9.661e-02  3.801e+03  -1.826  0.067984
## year_quarter2021Q4      -3.340e-01  8.956e-02  3.795e+03  -3.729  0.000195
## year_quarter2022Q1      -1.489e-01  8.941e-02  3.813e+03  -1.665  0.095926
## year_quarter2022Q2      -3.225e-01  8.735e-02  3.829e+03  -3.693  0.000225
## year_quarter2022Q3      -2.282e-01  8.998e-02  3.775e+03  -2.536  0.011241
## year_quarter2022Q4      -2.886e-01  1.099e-01  3.801e+03  -2.626  0.008668
## senate_tw               1.679e-02  6.502e-02  8.326e+02   0.258  0.796245
## republican_tw           -5.371e-02  6.483e-02  4.391e+02  -0.829  0.407832
##
## (Intercept)             ***
## emi_mean_floor          ***
## legislative_opposition_tw ***
## presidential_opposition_tw ***
## year_quarter2013Q2
## year_quarter2013Q3
## year_quarter2013Q4      .
## year_quarter2014Q1
## year_quarter2014Q2
## year_quarter2014Q3
## year_quarter2014Q4
## year_quarter2015Q1
## year_quarter2015Q2
## year_quarter2015Q3
## year_quarter2015Q4
## year_quarter2016Q1      ***
## year_quarter2016Q2      **
## year_quarter2016Q3      **
## year_quarter2016Q4
## year_quarter2017Q1      **
## year_quarter2017Q2
## year_quarter2017Q3      **
## year_quarter2017Q4      **
## year_quarter2018Q1
## year_quarter2018Q2
## year_quarter2018Q3
## year_quarter2018Q4
## year_quarter2019Q1
## year_quarter2019Q2      *
## year_quarter2019Q3
## year_quarter2019Q4
## year_quarter2020Q1      **
## year_quarter2020Q2      .
## year_quarter2020Q3
## year_quarter2020Q4
## year_quarter2021Q1      **
## year_quarter2021Q2
## year_quarter2021Q3      .
## year_quarter2021Q4      ***
## year_quarter2022Q1      .
## year_quarter2022Q2      ***
## year_quarter2022Q3      *
## year_quarter2022Q4      **
## senate_tw

```

```
## republican_tw
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation matrix not shown by default, as p = 45 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)          if you need it
```

```
vif(model_tw_floor)
```

```
##
##              GVIF Df GVIF^(1/(2*Df))
## emi_mean_floor      1.199950  1      1.095422
## legislative_opposition_tw 1.078188  1      1.038358
## presidential_opposition_tw 1.056884  1      1.028049
## year_quarter        1.132452 39      1.001596
## senate_tw           1.018954  1      1.009433
## republican_tw       1.015796  1      1.007867
```

```
# medium model
model_medium_base <- lmer(emi_sc ~ 1 + channel + year_quarter + (1|bioguide_id),
                          data=df_tw_congress
                          )
summary(model_medium_base)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: emi_sc ~ 1 + channel + year_quarter + (1 | bioguide_id)
## Data: df_tw_congress
##
## REML criterion at convergence: 20521.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.3790 -0.6152 -0.0155  0.6231  4.5199
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## bioguide_id (Intercept) 0.4254   0.6522
## Residual              0.5903   0.7683
## Number of obs: 8382, groups: bioguide_id, 467
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    2.807e-02  7.082e-02 5.276e+03  0.396 0.691837
## channelTwitter -1.499e-01  1.678e-02 7.862e+03 -8.931 < 2e-16 ***
## year_quarter2013Q2 -4.066e-02  7.712e-02 8.039e+03 -0.527 0.598068
## year_quarter2013Q3 -6.612e-02  8.194e-02 7.974e+03 -0.807 0.419705
## year_quarter2013Q4 -1.321e-01  8.268e-02 7.968e+03 -1.598 0.110136
## year_quarter2014Q1  8.882e-02  8.439e-02 7.998e+03  1.053 0.292601
## year_quarter2014Q2  2.334e-01  8.005e-02 7.973e+03  2.916 0.003559 **
## year_quarter2014Q3  1.651e-01  8.324e-02 7.975e+03  1.984 0.047310 *
```

```

## year_quarter2014Q4 7.725e-02 1.019e-01 7.951e+03 0.758 0.448292
## year_quarter2015Q1 -4.302e-02 8.069e-02 7.962e+03 -0.533 0.593920
## year_quarter2015Q2 1.418e-01 7.729e-02 8.037e+03 1.835 0.066517 .
## year_quarter2015Q3 4.862e-03 8.363e-02 7.969e+03 0.058 0.953643
## year_quarter2015Q4 1.159e-01 7.859e-02 7.980e+03 1.475 0.140199
## year_quarter2016Q1 3.323e-01 8.287e-02 8.013e+03 4.010 6.12e-05 ***
## year_quarter2016Q2 2.511e-01 7.717e-02 7.980e+03 3.254 0.001141 **
## year_quarter2016Q3 3.249e-01 8.257e-02 8.036e+03 3.935 8.40e-05 ***
## year_quarter2016Q4 4.602e-01 1.235e-01 7.924e+03 3.726 0.000196 ***
## year_quarter2017Q1 -2.521e-01 7.678e-02 8.009e+03 -3.283 0.001031 **
## year_quarter2017Q2 4.892e-02 8.454e-02 7.974e+03 0.579 0.562845
## year_quarter2017Q3 1.811e-01 7.935e-02 8.020e+03 2.282 0.022492 *
## year_quarter2017Q4 -7.713e-02 7.744e-02 8.016e+03 -0.996 0.319279
## year_quarter2018Q1 7.886e-02 8.223e-02 7.982e+03 0.959 0.337583
## year_quarter2018Q2 2.143e-01 7.780e-02 8.052e+03 2.754 0.005898 **
## year_quarter2018Q3 2.225e-01 8.841e-02 7.977e+03 2.516 0.011878 *
## year_quarter2018Q4 5.718e-02 9.987e-02 7.936e+03 0.573 0.566944
## year_quarter2019Q1 1.670e-02 7.996e-02 8.018e+03 0.209 0.834565
## year_quarter2019Q2 3.514e-01 7.961e-02 8.097e+03 4.414 1.03e-05 ***
## year_quarter2019Q3 4.243e-01 9.168e-02 8.101e+03 4.627 3.76e-06 ***
## year_quarter2019Q4 1.883e-01 7.844e-02 8.063e+03 2.401 0.016370 *
## year_quarter2020Q1 3.333e-01 8.409e-02 8.061e+03 3.963 7.45e-05 ***
## year_quarter2020Q2 3.148e-01 1.003e-01 7.997e+03 3.139 0.001701 **
## year_quarter2020Q3 1.213e-01 8.442e-02 8.069e+03 1.437 0.150694
## year_quarter2020Q4 1.769e-01 8.997e-02 8.035e+03 1.966 0.049284 *
## year_quarter2021Q1 -2.383e-01 8.436e-02 8.054e+03 -2.825 0.004737 **
## year_quarter2021Q2 2.232e-01 8.221e-02 8.144e+03 2.715 0.006651 **
## year_quarter2021Q3 -1.000e-01 9.149e-02 8.091e+03 -1.093 0.274420
## year_quarter2021Q4 -1.487e-01 8.480e-02 8.083e+03 -1.754 0.079505 .
## year_quarter2022Q1 -7.272e-02 8.469e-02 8.114e+03 -0.859 0.390502
## year_quarter2022Q2 9.262e-02 8.264e-02 8.136e+03 1.121 0.262439
## year_quarter2022Q3 2.387e-02 8.515e-02 8.048e+03 0.280 0.779277
## year_quarter2022Q4 1.449e-01 1.038e-01 8.088e+03 1.396 0.162747
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 41 > 12.
## Use print(x, correlation=TRUE) or
##     vcov(x)         if you need it

model_medium <- lmer(emi_sc ~ 1 + channel + legislative_opposition
                     + presidential_opposition
                     + year_quarter + senate
                     + republican + (1|bioguide_id),
                     data=df_tw_congress
                     )
summary(model_medium)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## emi_sc ~ 1 + channel + legislative_opposition + presidential_opposition +

```

```

##      year_quarter + senate + republican + (1 | bioguide_id)
##      Data: df_tw_congress
##
## REML criterion at convergence: 19494.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.7459 -0.6156 -0.0036  0.6009  5.3737
##
## Random effects:
##      Groups          Name          Variance Std.Dev.
## bioguide_id (Intercept) 0.3699    0.6082
## Residual                0.5213    0.7220
## Number of obs: 8382, groups: bioguide_id, 467
##
## Fixed effects:
##
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    6.385e-01  7.559e-02  3.071e+03   8.446 < 2e-16 ***
## channelTwitter -1.499e-01  1.577e-02  7.865e+03  -9.504 < 2e-16 ***
## legislative_opposition -4.338e-01  1.985e-02  8.334e+03 -21.857 < 2e-16 ***
## presidential_opposition -4.159e-01  1.745e-02  8.213e+03 -23.834 < 2e-16 ***
## year_quarter2013Q2    -6.325e-02  7.251e-02  8.041e+03  -0.872 0.383062
## year_quarter2013Q3    -6.679e-02  7.703e-02  7.978e+03  -0.867 0.385912
## year_quarter2013Q4    -1.641e-01  7.771e-02  7.969e+03  -2.112 0.034731 *
## year_quarter2014Q1     6.111e-02  7.931e-02  8.000e+03   0.771 0.441008
## year_quarter2014Q2     2.200e-01  7.525e-02  7.977e+03   2.924 0.003467 **
## year_quarter2014Q3     1.720e-01  7.825e-02  7.980e+03   2.198 0.027951 *
## year_quarter2014Q4     6.629e-02  9.574e-02  7.954e+03   0.692 0.488717
## year_quarter2015Q1    -5.902e-02  7.583e-02  7.964e+03  -0.778 0.436355
## year_quarter2015Q2     1.319e-01  7.264e-02  8.036e+03   1.816 0.069411 .
## year_quarter2015Q3    -1.129e-05  7.859e-02  7.971e+03   0.000 0.999885
## year_quarter2015Q4     1.180e-01  7.388e-02  7.982e+03   1.598 0.110129
## year_quarter2016Q1     3.151e-01  7.789e-02  8.015e+03   4.046 5.26e-05 ***
## year_quarter2016Q2     2.605e-01  7.253e-02  7.983e+03   3.591 0.000332 ***
## year_quarter2016Q3     3.349e-01  7.761e-02  8.039e+03   4.315 1.61e-05 ***
## year_quarter2016Q4     4.809e-01  1.161e-01  7.928e+03   4.142 3.47e-05 ***
## year_quarter2017Q1    -1.612e-01  7.226e-02  8.014e+03  -2.231 0.025704 *
## year_quarter2017Q2     1.305e-01  7.952e-02  7.983e+03   1.641 0.100899
## year_quarter2017Q3     2.477e-01  7.465e-02  8.023e+03   3.319 0.000909 ***
## year_quarter2017Q4    -2.703e-02  7.283e-02  8.023e+03  -0.371 0.710541
## year_quarter2018Q1     9.648e-02  7.730e-02  7.987e+03   1.248 0.212016
## year_quarter2018Q2     2.423e-01  7.314e-02  8.054e+03   3.313 0.000927 ***
## year_quarter2018Q3     2.605e-01  8.313e-02  7.983e+03   3.133 0.001737 **
## year_quarter2018Q4     8.352e-02  9.389e-02  7.940e+03   0.890 0.373724
## year_quarter2019Q1     3.881e-02  7.522e-02  8.023e+03   0.516 0.605843
## year_quarter2019Q2     3.697e-01  7.490e-02  8.095e+03   4.936 8.12e-07 ***
## year_quarter2019Q3     4.254e-01  8.624e-02  8.104e+03   4.932 8.30e-07 ***
## year_quarter2019Q4     1.981e-01  7.377e-02  8.069e+03   2.686 0.007249 **
## year_quarter2020Q1     3.326e-01  7.907e-02  8.068e+03   4.206 2.62e-05 ***
## year_quarter2020Q2     2.765e-01  9.431e-02  8.005e+03   2.932 0.003379 **
## year_quarter2020Q3     1.144e-01  7.937e-02  8.075e+03   1.441 0.149534
## year_quarter2020Q4     1.578e-01  8.457e-02  8.043e+03   1.866 0.062023 .
## year_quarter2021Q1    -2.672e-01  7.930e-02  8.059e+03  -3.369 0.000758 ***
## year_quarter2021Q2     2.227e-01  7.728e-02  8.147e+03   2.882 0.003968 **

```

```
## year_quarter2021Q3      -1.194e-01  8.600e-02  8.098e+03  -1.389  0.165016
## year_quarter2021Q4      -1.504e-01  7.973e-02  8.088e+03  -1.886  0.059283 .
## year_quarter2022Q1      -6.248e-02  7.960e-02  8.118e+03  -0.785  0.432518
## year_quarter2022Q2       5.448e-02  7.769e-02  8.137e+03   0.701  0.483157
## year_quarter2022Q3       1.603e-02  8.005e-02  8.051e+03   0.200  0.841287
## year_quarter2022Q4       1.976e-01  9.762e-02  8.090e+03   2.024  0.043028 *
## senate                  -2.464e-01  6.053e-02  9.509e+02  -4.070  5.08e-05 ***
## republican              -2.291e-01  6.168e-02  4.435e+02  -3.714  0.000230 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation matrix not shown by default, as p = 45 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)          if you need it
```

```
vif(model_medium)
```

```
##                GVIF Df GVIF^(1/(2*Df))
## channel                1.000000    1      1.000000
## legislative_opposition  1.011616    1      1.005791
## presidential_opposition 1.021966    1      1.010924
## year_quarter           1.039838   39      1.000501
## senate                 1.014107    1      1.007029
## republican             1.009885    1      1.004930
```

```
# twitter predicts floor
```

```
modelsummary(
  list(model_floor_tw_base, model_floor_tw),
  vcov = list(vcovCR(model_floor_tw_base, cluster = merged$bioguide_id, type = "CR2"),
             vcovCR(model_floor_tw, cluster = merged$bioguide_id, type = "CR2")
  ),
  coef_omit = "^bioguide_id|^year_quarter",
  gof_omit = "AIC|BIC|RMSE|Log.Lik.",
  output = "../output/Twitter-Congress-meff_summary.html",
  fmt = NULL,
  estimate = "{f_est(estimate)} [{f_est(conf.low)}, {f_est(conf.high)}]",
  statistic = "p={f_p(p.value)}", conf_level = 0.95,
  notes = "Values in square brackets represent 95% confidence intervals based on cluster-robust standard errors"
)
```

```
tex <- modelsummary(
  list(model_floor_tw_base, model_floor_tw),
  vcov = list(vcovCR(model_floor_tw_base, cluster = merged$bioguide_id, type = "CR2"),
             vcovCR(model_floor_tw, cluster = merged$bioguide_id, type = "CR2")
  ),
  coef_omit = "^bioguide_id|^year_quarter",
  gof_omit = "AIC|BIC|RMSE|Log.Lik.",
  output = "tinytable",
  fmt = NULL,
  estimate = "{f_est(estimate)}",
  statistic = c(
    "[{f_est(conf.low)}, {f_est(conf.high)}]",

```

```

      "p={f_p(p.value)}"
    ),
    #estimate = "{f_est(estimate)} [{f_est(conf.low)}, {f_est(conf.high)}]",
    #statistic = "p={f_p(p.value)}",
    conf_level=0.95,
    notes = "Values in square brackets represent 95% confidence intervals based on cluster-robust standard errors"
  )
)|>theme_tt("tabular")
  save_tt(tex, "../output/Twitter-Congress-meff_summary.tex", overwrite = TRUE)

# floor predicts twitter
modelsummary(
  list(model_tw_floor_base, model_tw_floor),
  vcov = list(vcovCR(model_tw_floor_base, cluster = merged$bioguide_id, type = "CR2"),
             vcovCR(model_tw_floor, cluster = merged$bioguide_id, type = "CR2")),
  coef_omit = "^bioguide_id|^year_quarter",
  gof_omit="AIC|BIC|RMSE|Log.Lik.",
  output = "../output/floor-predictor-meff_summary.html",
  fmt = NULL,
  estimate = "{f_est(estimate)} [{f_est(conf.low)}, {f_est(conf.high)}]",
  statistic = "p={f_p(p.value)}", conf_level=0.95,
  notes = "Values in square brackets represent 95% confidence intervals based on cluster-robust standard errors"
)

tex <- modelsummary(
  list(model_tw_floor_base, model_tw_floor),
  vcov = list(vcovCR(model_tw_floor_base, cluster = merged$bioguide_id, type = "CR2"),
             vcovCR(model_tw_floor, cluster = merged$bioguide_id, type = "CR2")),
  coef_omit = "^bioguide_id|^year_quarter",
  gof_omit="AIC|BIC|RMSE|Log.Lik.",
  output = "tinytable",
  fmt = NULL,
  estimate = "{f_est(estimate)}",
  statistic = c(
    "[{f_est(conf.low)}, {f_est(conf.high)}]",
    "p={f_p(p.value)}"
  ),
  conf_level=0.95,
  notes = "Values in square brackets represent 95% confidence intervals based on cluster-robust standard errors"
)|>theme_tt("tabular")
  save_tt(tex, "../output/floor-predictor-meff_summary.tex", overwrite = TRUE)

# medium model
modelsummary(
  list(model_medium_base, model_medium),
  vcov = list(vcovCR(model_medium_base, cluster = df_tw_congress$bioguide_id, type = "CR2"),
             vcovCR(model_medium, cluster = df_tw_congress$bioguide_id, type = "CR2")),
  coef_omit = "^bioguide_id|^year_quarter",
  gof_omit="AIC|BIC|RMSE|Log.Lik.",
  output = "../output/Twitter-Congress-medium_summary.html",
  fmt = NULL,

```

```

estimate = "{f_est(estimate)} [{f_est(conf.low)}, {f_est(conf.high)}]",
statistic = "p={f_p(p.value)}", conf_level=0.95,
notes = "Values in square brackets represent 95% confidence intervals based on cluster-robust standard errors"
)

tex <- modelsummary(
  list(model_medium_base, model_medium),
  vcov = list(vcovCR(model_medium_base, cluster = df_tw_congress$bioguide_id, type = "CR2"),
    vcovCR(model_medium, cluster = df_tw_congress$bioguide_id, type = "CR2")
  ),
  coef_omit = "^bioguide_id|^year_quarter",
  gof_omit = "AIC|BIC|RMSE|Log.Lik.",
  output = "tinytable",
  fmt = NULL,
  estimate = "{f_est(estimate)}",
  statistic = c(
    "[{f_est(conf.low)}, {f_est(conf.high)}]",
    "p={f_p(p.value)}"
  ),
  conf_level=0.95,
  notes = "Values in square brackets represent 95% confidence intervals based on cluster-robust standard errors"
)|>theme_tt("tabular")
save_tt(tex, "../output/Twitter-Congress-medium_summary.tex", overwrite = TRUE)

```